

**UNIVERSITY OF TECHNOLOGY
(YATANARPON CYBER CITY)
FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY
DEPARTMENT OF INFORMATION SCIENCE**

**IMMERSIVE LEARNING THROUGH AUGMENTED
REALITY FOR CHILDREN'S QUALITY EDUCATION**

**BY
THWIN HTOO AUNG**

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THWIN HTOO AUNG

A partial dissertation submitted to the University of Technology (Yatanarpon Cyber City) in
fulfillment of the requirement for the degree of

Bachelor of Engineering
(Information Science and Technology)

August, 2024

DECLARATION

I hereby declare that I carried out the work reported in this thesis in the Department of Information Science under the Faculty of Information and Communication Technology, University of Technology (Yatanarpon Cyber City), under the supervision of Dr. May Mar Oo. I solemnly declare that to the best of my knowledge, no part of this thesis has been submitted here or elsewhere in a previous application for the award of a degree. All sources of knowledge used have been duly acknowledged.

.....

27th August, 2024

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APPROVAL

This is to certify that the thesis titled “**Immersive Learning through Augmented Reality for Children’s Quality Education**” carried out by **Thwin Htoo Aung, 6IST-1** has been read and approved for meeting part of the requirements and regulations governing the award of the degree of Bachelor of Engineering (Information Science and Technology), Department of Information Science under the Faculty of Information and Communication Technology, University of Technology (Yatanarpon Cyber City), Myanmar.

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ABSTRACT

This system is about an Augmented Reality (AR) application with simulated 3D models where children can study in 360°degree immersive and interactive learning experience. This application offers opportunities to revolutionize educational methodologies and improve learning outcomes, particularly for children. It is designed as a teaching aid for shaping and enhancing the quality of education. This system will introduce the concept of Metaverse technology (Augmented Reality (AR), Virtual Reality (VR), Mixed Reality (MR) and Extended Reality (XR)) and their applications in educational sector, especially my focus here, Augmented Reality (AR). This AR application will be built with Unity Software Development and Vuforia Engine which will detect and track image target by comparing natural features from the camera image against a known target resource database. This application offers a captivating learning environment where children can study English vocabularies with interactive and entertaining 3D models, improving kid's vocabulary skills.

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LIST OF ABBREVIATION AND SYMBOLS

Abbreviation/Symbol	Description
AR	Augmented Reality
VR	Virtual Reality
MR	Mixed Reality
XR	Extended Reality
IoT	Internet of Things
VST	Video See-through
OST	Optical See-through
RGB	Red-Green-Blue
SIFT	Scale-Invariant Feature Transform
3D	Three-dimensional

CHAPTER 1

INTRODUCTION

Access to quality education is universally recognized as a fundamental right and a cornerstone of individual empowerment and societal development. However, in many regions, including Myanmar, ensuring equitable access to quality education remains a significant challenge due to factors such as limited teaching resources, inadequate infrastructure, and teacher shortages. Traditional teaching methods often fall short in meeting the diverse learning needs of children, particularly in resource-constrained environments.

Conventional teaching methodologies, while valuable, often struggle to engage students fully and cater to their diverse learning needs. As such, there is a growing imperative to explore innovative approaches that can revolutionize education and unlock the full potential of every child.

In the rapidly evolving landscape of education, the integration of technology has become pivotal in reshaping traditional pedagogical approaches and enriching learning experiences. The emergence of Metaverse technology, including AR, VR, MR, and XR, presents an opportunity to overcome these challenges and revolutionize learning.

This system proposes a solution by utilizing AR technology as a teaching aid to enhance traditional teaching methods and elevate the quality of education for children in Myanmar. Among the various manifestations of Metaverse technology, Augmented Reality stands out as a particularly promising tool for transforming education. By overlaying digital content onto the physical world, AR bridges the gap between the virtual and the real, enabling learners to interact with digital objects and information in their immediate environment. This system specifically focuses on improving English vocabulary skills through an AR 360-degree app developed using Unity and Vuforia Engine.

This system explores the implementation of an Augmented Reality (AR) application featuring simulated 3D models to provide children with a 360-degree immersive and interactive learning experience. The application aims to revolutionize educational methodologies and enhance learning outcomes, particularly for children,

serving as a valuable teaching aid. Focused on the educational sector, specifically Augmented Reality (AR), this system introduces the concept of Metaverse technology (including Augmented Reality (AR), Virtual Reality (VR) and Mixed Reality (MR)) and its potential applications.

The AR application is developed using Unity Software Development and Vuforia Engine, utilizing image target detection and tracking to create a captivating learning environment where children can improve their English vocabulary skills through interactive 3D models.

This system includes a vocabulary dataset comprising English vocabulary flashcard images and a 3D model dataset associated with these images. Additionally, there is an audio dataset for pronunciation, virtual 360-degree viewing angle feature, and control of the 3D models. Image targets are created using the Vuforia Target Manager with JPG or PNG images, with specific size and resolution requirements.

1.1 Objectives

The objectives of this system are as follows:

- To support “Quality Education” for all children with the integration of AR technology
- To provide as a teaching aid, addressing the challenge of limited teaching resources in Myanmar
- To implement an AR 360°degree application, enhancing vocabulary skills for children
- To be an inspiration for future Metaverse projects

1.2 Application Background

In today's rapidly evolving educational landscape, the integration of technology has become crucial in reshaping traditional pedagogical methods and enhancing learning experiences. The emergence of Metaverse technologies, which include Augmented Reality (AR), Virtual Reality (VR), Mixed Reality (MR), and Extended Reality (XR), presents a unique opportunity to overcome these challenges and transform learning processes.

This system proposes a solution by utilizing AR technology as a teaching aid to enhance traditional educational methods and elevate the quality of education for

children in Myanmar. Among the various manifestations of Metaverse technology, AR stands out as a particularly promising tool for transforming education. By overlaying digital content onto the physical world, AR bridges the gap between the virtual and the real, allowing learners to interact with digital objects and information within their immediate environment.

This system introduces “AR-KID,” an AR-based application designed to enhance English vocabulary skills for children in Myanmar. Developed using Unity and Vuforia, AR-KID provides an immersive 360-degree learning environment with three main features:

1. 3D AR Session: Users scan A-Z flashcards, triggering 3D models (e.g., an alligator for "A"), making vocabulary learning engaging and memorable.

2. AR Mag Feature: Scanning magazine papers on topics like the Solar System and cities around the world renders associated 3D objects and live-motion videos, enriching the learning experience.

3. Eng. Poem Feature: Scanning flashcards linked to popular children’s poems triggers a virtual video player, allowing users to watch and interact with the videos.

AR-KID is designed to be scalable and extensible, ensuring it can be updated with new content to meet evolving educational needs. This application exemplifies how AR technology can revolutionize traditional education by making it more interactive and effective.

1.3 Overview of the System

The system developed for this thesis is an Augmented Reality (AR) application designed to provide children with a captivating and immersive learning experience. Leveraging the Unity Software Development platform and the Vuforia Engine, the application integrates cutting-edge AR technology to create a dynamic educational environment.

This AR application is named as “AR-KID”. “AR-KID” application will have three main features- 3D AR, AR Mag and Eng. Poem. In 3D AR session, the users can scan twenty-six A-Z flashcards and this application will show their associated 3D models. For example, if they scan “A” flashcard for Alligator, this app will render 3D model of alligator that looks like popping up from this flashcard. In AR Mag feature, there will be five magazine papers with the topics of “Solar System”, “Cities Around the World”, “Sport Mag”, “Computer System and Peripherals” and “Missing Dog”.

When my users scan these five magazine papers, and their associated multiple 3D objects and live-motion videos will be rendered. In the last one feature -Eng. Poem, four English poem flashcards are available. These are “Baby Shark”, “Twinkle Twinkle Little Star”, “Baa Baa Black Sheep” and “ABCD Poem”. When the users scan these poem flashcards, their virtual video player will pop up and they can enjoy the videos and control video player.

The system is implemented using the Unity Software Development platform, which provides a robust framework for creating cross-platform AR applications. The Vuforia Engine is utilized for image target recognition and tracking, enabling seamless integration of digital content into the real world. Furthermore, the system is designed with scalability and extensibility in mind, allowing for future updates and enhancements to be easily incorporated.

1.4 Organization of the Thesis

The primary function of the system is to facilitate vocabulary learning for children. Through interactive 3D models and accompanying textual descriptions, users can explore and learn new words in an engaging and interactive manner.

This thesis is structured as follows:

Chapter 1 describes an introduction to the topic, objective of the study, field background and the overview of the system.

Chapter 2 presents the theoretical background in details. It describes what the metaverse is, and discuss about Virtual Reality (VR), Mixed Reality (MR) and especially Augmented Reality (AR). It presents Augmented Reality development for Quality Education, digital image processing techniques and app development with Vuforia engine in Unity. And then, it explains about image feature extraction process by using SIFT (Scale Invariant Feature Transform) algorithm and 3D modeling.

Chapter 3 provides the system design, system flow, the design and implementation of the system and user interface design of the system.

Chapter 4 expresses the conclusion, benefits of the system, limitations of the system and further extension.

CHAPTER 2

THEORETICAL BACKGROUND

This chapter includes the detail of theoretical background that inform and contextualize the development and implementation of the Augmented Reality (AR) application of Metaverse technology for immersive learning in children's education. Metaverse is an upcoming transformative technology that will impact our future society with immersive experiences. The recent surge in the adoption of new technologies and innovations in connectivity, interaction technology, and artificial realities can fundamentally change the digital world. The Metaverse concept is the most recent trend to encapsulate and define the potential new digital landscape. However, with the introduction of 5G with high speed and low latency advancements in the hardware and software with the graphics power to display millions of polygons in 3D and blockchain technology, this concept is no longer fiction. This transition from today's Internet to a spatially embodied Internet is, at its core, a transition from 2D to 3D interactions taking place in multiple virtual universes. In recent years, augmented virtual reality has created possibilities in the private and professional spheres. The new Virtual Reality (VR) headsets and Augmented Reality (AR) glasses can provide immersion in the physical sense. Technology must offer realistic experiences for users to turn this concept into reality. This system focuses on relevant literature on immersive learning, augmented reality, and their applications in the educational domain.

2.1 What is Metaverse?

Imagine a new universe free from the physical realities of your life where you can work, connect with people, and relax anywhere you want on the planet. With opportunities to attend gatherings and do anything that you can imagine in reality, this dream is not far. Now it is the beginning of the Metaverse as the next evolutionary [3].

Metaverse is a compound word combined with “meta-” (beyond; transcending) and “verse” (the root of “universe,” cosmos; the whole world), which denotes a new virtual universe created beyond the real world. The term “metaverse” was first coined in the 1992 cyberpunk science fiction novel *Snow Crash* written by American novelist

Neal Stephenson. In this novel, humans could freely access a 3D space that reflects the real world through digital agents (avatars) and interact with each other. Over the next three decades, the metaverse concept was more vividly depicted in science fiction movies, such as Ready Player One, Lucy, and The Matrix. At that time, the metaverse envisioned by films, could not come into being in reality. In this decade, the rapid progress of emerging technologies, such as wearable devices and three-dimensional (3D) photography, has helped people to get access to the virtual world. This sleeping “lion” was truly awakened.

The metaverse is not merely a new entity for VR or AR, but a collection of a set of emergent technologies like 5G, AI, VR, AR, digitally twins, blockchain, or IoT (Internet of Things). The technological framework might be built for specific realms such as entertainment, commerce, education, etc., and their components and functions can be different according to the needs. With the support of high-speed networks such as 5G/6G, users can use smart wearable devices (e.g., headsets or glasses) to enter the metaverse world instantly without being constrained by either time or location.

AR overlays digital information onto the real-world environment to enhance the real-world perception with digital elements, VR creates entirely immerses computer-generated environments by blocking out the physical world, while Mixed Reality (MR) enables users to interact with digital and physical objects simultaneously by blending digital content into the real world.

Instead of a static image, in the metaverse, each user could customize his/her unique digital identity in the form of an avatar. The construction of the digital identity is more user-defined and more advanced than before, for instance, it could edit the details of the avatar’s face, body, and even facial expression. It is the surrogate identity of users in the virtual world which reflects the user’s persona and represents the ego in the real world. In addition, avatars can be manipulated and controlled by users with the help of real-time tracking technologies. In this case, the living 3D representation of users (i.e., digital identity) plays an important role in ownership, interactivity, embodiment, and socialization in the metaverse world.

Metaverse is also used to describe gaming worlds where every user can have a virtual character to interact with other players’ virtual characters. Taking the VR from gaming to this virtual world for practically everything, Metaverse will be for work, education, healthcare, conferences, play, concerts, or just hanging out. The concept of the Metaverse holds significant potential for positive societal impact and “tech for

good” applications. “Tech for good” is a concept that refers to the use of technology and innovation to create positive and beneficial outcomes for individuals, communities, and society as a whole. It involves leveraging technological advancements to address social, environmental, and humanitarian challenges, with the aim of improving people’s lives and promoting well-being. “Tech for good” initiatives focus on developing and deploying technological solutions that contribute to sustainable development, equality, inclusivity, and positive societal impact.

Across the globe, AR is becoming an integral part of the educational systems that will move forward to the Metaverse. Figure 2.1 explains the educational applications of Metaverse. Education has rich traditional practices of classrooms, curriculum structures, grading systems, etc., but so far in the digital transformation, it is impossible to turn it into virtual spaces completely. This need emerged during the COVID pandemic when the world felt the need for virtual classrooms.

As an immersive virtual space, the Metaverse concept can bring the campus activities and class lectures with 3D simulations to enable a more realistic classroom learning experience where everyone presents virtually in an immersive form. With the involvement of activity-oriented and enhanced gamification, it can increase classroom participation.

Metaverse’s applications in education include, but are not limited to, virtual 3D classrooms, simulated real-life situations, digital learning in a remote setting, interdisciplinary learning, virtual campus tours, events, and collaborative activities. It can help to create powerful immersive learning experiences, gamification, hands-on learning practices, and improve learning speed.

By bringing the concept of social presence, Metaverse can help students to meet and interact virtually with other students and teachers in 3D virtual classrooms. Metaverse’s ability to replicate real-life situations where students can conduct scientific experiments can revolutionize STEM learning, especially in resource-constrained environments.

A prototype of a campus Metaverse at the Chinese University of Hong Kong, Shenzhen was presented with a blockchain-driven system (explained in the following Figure 2.2). The system aims to provide an on-campus interactive Metaverse for students with a mixed environment where students’ real-world actions reflect in the virtual world and showcase different categories of Metaverse’s applications in education sector around the world.

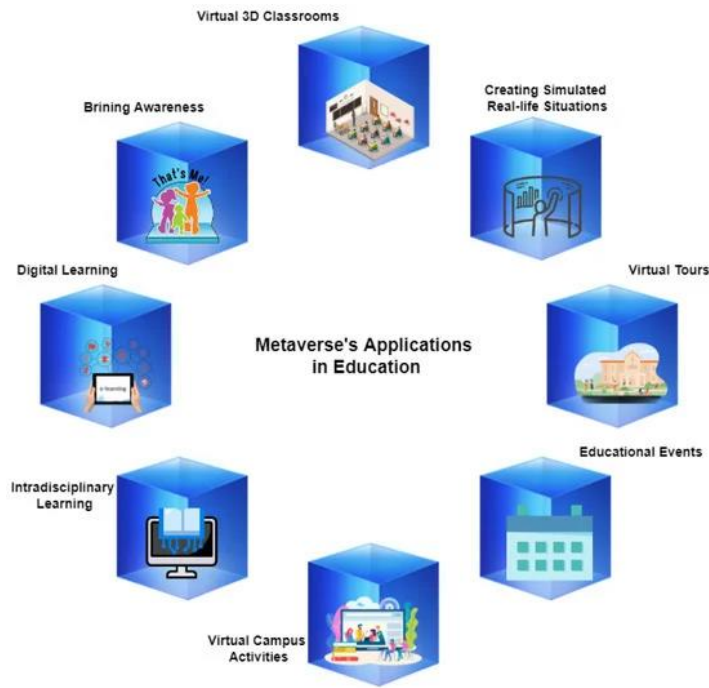


Figure 2.1: Metaverse’s Applications in Education



Figure 2.2: A Corner of Chinese University of Hong Kong, Shenzhen as Metaverse

The exciting thing about the Metaverse in the context of education is, learning seems to be a perfect fit as it will empower learning by doing “kinesthetic learning” at its best with simulated activities about experiential learning. In 2021, Stanford University started its course COMM 166/266: “Virtual People” completely based on virtual reality. The evaluation of a collaborative e-learning Metaverse platform, VoRtex shows positive results in overcoming the learning boundaries. A bottom-up approach adopted for real case studies to use Metaverse-based education has been shown to create Sustainable Learning experiences. Learning technical topics and training in remote and resource-constrained environments is possible using artificial technologies, and the

concept of haptic feedback can take this approach to the next level. Using haptic sensing gloves would allow interaction with the virtual world and experience it in the same way as we do in the real world. In artificial reality, the goal of haptic technology is comfortable and lightweight gloves to convey tactile information such as pressure, heat, and weight. Suzuki et al. used the concept for the learning system in Metaverse. Figure 2.3 shows the Meta haptic gloves for technical training.



Figure 2.3. Haptic Feedback Gloves Revealed by Facebook Reality Labs

Traditional learning resources generally exist in relatively static forms, such as printed textbooks, printed papers, electronic books, shared courseware, pictures, videos, or other materials. Learners are less likely to interact with those learning resources. In metaverse-based education, the learning resources are visualized and decentralized which enables learners to interact. Take a lesson on the topic of the earth, for example, there may be a lecture about the earth with a printed textbook and a demonstration using physical objects of a terrestrial globe and a map in the traditional teaching session; while with the help of AR, the learning resources can be entirely different than before: a 3D spinning earth modeled and augmented by technologies. Learners can observe the virtual earth at 360° by zooming in, zooming out, and rotating it. Hence, in the metaverse, some abstract content can be turned into more concrete by modeling and rendering, thereby enhancing learners' in-depth understanding. It is also possible for learners to interact with learning resources to facilitate their participation and experience, as well as participate in the process of creating or editing learning resources along with their peers and teachers, which may compensate for certain shortcomings associated with traditional resources [3].

2.1.1 Augmented Reality

Augmented Reality (AR) is a technology that superimposes digital information, such as images, videos, or 3D models, onto the user's view of the real world. Unlike Virtual Reality (VR), which immerses users in a completely virtual environment, AR enhances the real-world environment by adding virtual elements. This integration of digital and physical worlds creates immersive experiences that have numerous applications across industries, including education [9].

AR can be defined a newer technological system in which virtual objects are added to the real world in real-time during the user's experience. An AR system should: (1) combine real and virtual objects in a real environment; (2) run interactively and in real-time; (3) register real and virtual objects with each other. Furthermore, even if the AR experiences could seem different from VRs, the quality of AR experience could be considered similarly. Indeed, like in VR, feeling of presence, level of realism, and the degree of reality represent the main features that can be considered the indicators of the quality of AR experiences. Higher the experience is perceived as realistic, and there is congruence between the user's expectation and the interaction inside the AR environments, higher would be the perception of "being there" physically, and at cognitive and emotional level. The feeling of presence, both in AR and VR environments, is important in acting behaviors like the real ones.

Technologically, the AR systems, however various, present three common components, such as a geospatial datum for the virtual object, like a visual marker, a surface to project virtual elements to the user, and an adequate processing power for graphics, animation, and merging of images, like a pc and a monitor. To run, an AR system must also include a camera able to track the user movement for merging the virtual objects, and a visual display, like glasses through that the user can see the virtual objects overlaying to the physical world. To date, two-display systems exist, a video see-through (VST) and an optical see-through (OST) AR systems. The first one, disclosures virtual objects to the user by capturing the real objects/scenes with a camera and overlaying virtual objects, projecting them on a video or a monitor, while the second one, merges the virtual object on a transparent surface, like glasses, through the user see the added elements. The main difference between the two systems is the latency: an OST system could require more time to display the virtual objects than a VST system, generating a time lag between user's action and performance and the detection of them by the system.

In education, several AR applications have been developed in the last few years showing the positive effects of this technology in supporting learning, such as an increased-on content understanding and memory preservation, as well as on learning motivation. For example, Ibanex Etal developed an AR application on electro-magnetism concepts' learning, in which students could use AR batteries, magnets, cables on real superficies, and the system gave real-time feedback to students about the correctness of the performance, improving in this way the academic success and motivation. Deeply, AR system allows the possibility to learn visualizing and acting on composite phenomena that traditionally students study theoretically, without the possibility to see and test in real world.

2.1.2 Virtual Reality

Virtual Reality (VR) is a technology that immerses users in a simulated environment that is entirely computer-generated. By wearing VR headsets or goggles, users are transported to virtual worlds where they can interact with objects and environments in three dimensions. VR provides a fully immersive experience for applications such as gaming, simulations, and virtual tours [10].

Although AR and VR might sound similar, but they are two distinct technologies with different purposes. VR immerses users in a completely new digital environment, providing an interactive experience through the use of headsets or glasses. AR complements the real-world environment by overlaying digital objects onto it, augmenting it with extra information or enhancing its functionality.

The key to VR is its ability to fully immerse the user in a digital environment, making it a powerful tool for creating experiences that feel real, even though they are entirely computer-generated.

2.1.3 Mixed Reality

Mixed reality is a hybrid of VR and AR and aims to offer the best of both worlds. For instance, while it uses a headset just like VR, seeing through a translucent viewport or glass, it also projects visuals on top of our environment.

In Mixed Reality, users can see and engage with both the physical environment around them and virtual objects simultaneously. This technology enables users to manipulate and control virtual elements as if they were part of the real world. For

example, with MR, a user could place a digital 3D model on a real table and interact with it, rotating or moving it as if it were a physical object.

MR blends physical and digital realities to create environments where virtual objects appear to exist in the user's physical space. Figure 2.4 shows the difference in Virtual Reality (VR), Augmented Reality (AR), Mixed Reality (MR) [10].

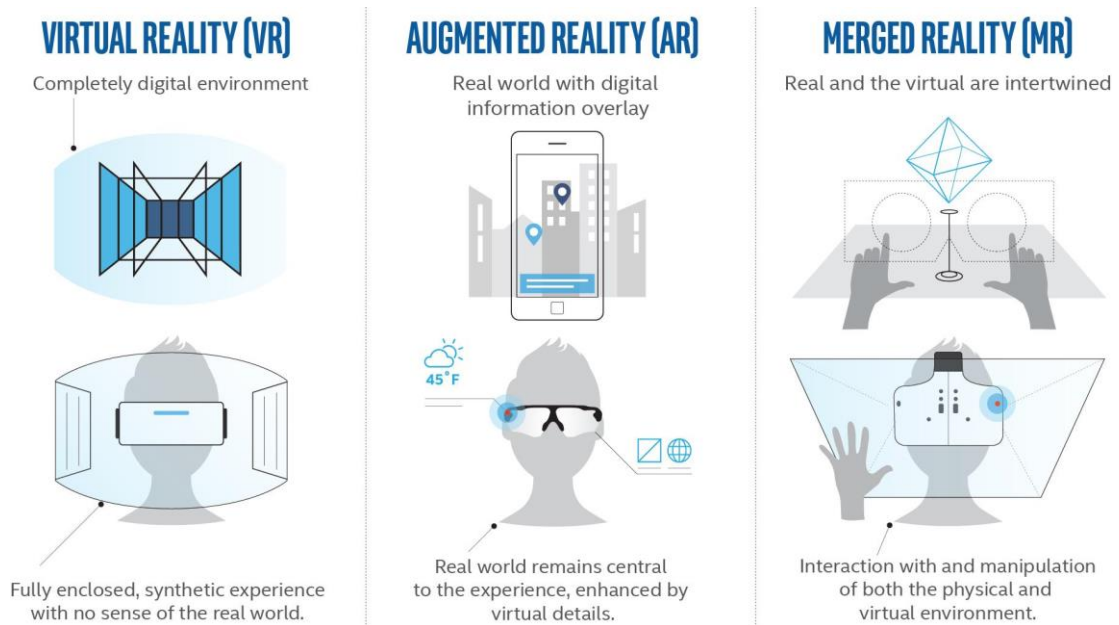


Figure 2.4: Virtual Reality (VR), Augmented Reality (AR), Mixed Reality (MR).

According to the main purpose of this system, reviewing existing literature during last 10 years of VR and AR in education and its recent developments and directions, related works have been summarized in the same area including multiple objectives and investigating specific usage of VR and AR in learning have been reviewed in Table 2.1.

Table 2.1. Related Works to VR & AR in Education in Recent Years.

Reference	Subject	Method	Findings
Zwoliński et al. (2022)	Extended reality in management education	Case Studies	Creating a modal for XR-based educational environment by utilizing different XR technologies
Scavarelli et al. (2021)	VR and AR in Social Learning	Literature Review	Exploring the recent developments of VR & AR in Social space and several learning theories.
Patel et al. (2020)	VR, AR & mixed Reality in education	Survey	Overview VR, AR and Mixed Reality in education and ability of people adaptation of these technologies.

Boyles (2017)	VR & AR in Education	Review	Describing usage of AR & AR in enhancing learning and reviewing advantages and disadvantages.
Olbina and Glick (2022).	Integration of AR & VR in Construction Management	Physical model	Improving visualization, improving in understanding of construction material and improving student communications skills.
Sirohi et al. (2020)	Augmented & Virtual Reality applications	Survey	An interdisciplinary review of integration of VR & AR in different area and directions.
Huang et al. (2019)	AR & VR in Education	Exploratory Study	Virtual reality is more inclusive of spatial presence while augmented reality is more effective in dealing with auditory information
Guo et al. (2021)	Extended Reality (XR) in Education	bibliometric analysis	Exploring the overall productivity of XR and recent development and trends in educational field.
Solmaz and Van Gerven (2022)	Integration of extract-based CFD of AR & VR	A architecture Modal	Proposing a component-oriented system architecture and Data simulation with AR/VR.
Nguyen and Dang (2017, October)	Setting up VR & AR learning environment	Designing Model	Designing 3D framework for curriculum based on VR &AR presetting real world objects
Cieri et al (2021)	Visual and Augmented Reality	Descriptive & case studies	Describing recent VR/AR software, hardware and techniques in pedagogy.
Rau et al. (2018)	Speed reading on VR & AR	Survey	Exploring performance on VR & AR where response time was longer on VR and AR than desktop
Remolar et al. (2021)	Learning throughout VR & AR	Several Experiments	Supporting gameplay and attractiveness and increasing student's interest to learn.
Beck (2019)	Immersive learning	Special issue	Integration of technologies including AR & VR to enhance learning environments.

2.2 Augmented Reality Development for Quality Education

Every child has the fundamental right to quality education. In Myanmar, ensuring access to quality education for children remains a biggest challenge, caused

by factors such as limited number of teaching aid resources, insufficient infrastructure, and teacher shortages. Quality education is not only essential for individual citizen empowerment but also plays a crucial role in the development of a country. However, traditional teaching methods is not enough in meeting the diverse learning needs of children, particularly in resource-constrained environments. In response to these challenges, the emergence of the Metaverse technology, comprising Augmented Reality (AR), Virtual Reality (VR), Mixed Reality (MR) and Extended Reality (XR), represents a unique opportunity to solve educational challenges, shaping the future of learning for children [11].

This system proposes a solution by using Augmented Reality (AR) technology as a teaching aid to revolutionize traditional teaching methods and upgrade the quality of education for Myanmar children. By using Unity software development and Vuforia Engine, this study aims to develop an AR 360°degree app specifically designed to improve children's English vocabulary skills, engaging with simulated 3D models in a 360°-degree immersive and interactive learning experience.

As of today, AR only requires a smartphone with a camera and an AR app. Two key elements that make it work are the camera capacity to capture the environment around you as you move and the software that calculates and projects some computer-generated visuals or content. Developing AR application includes concepts such as image detection and tracking, spatial mapping, and rendering virtual content in real-world environments.

This system titled “Immersive Learning through Augmented Reality for Children’s Quality Education” includes vocabulary dataset in which a collection of English vocabularies flashcard images and 3D Model dataset where their associated 3D models will be pop up in the surface of these image target.

There will also be audio dataset about pronunciation, virtual 360°degree viewing angle feature, and the control of these 3D models. Image targets will be created with the Vuforia Target Manager using JPG or PNG images in RGB or grayscale. The size of the input images must be 2.25 MB or less and a minimum width of 320 pixel.

2.3 Digital Image Processing

Digital image processing technology is a method to transform image signals into digital signals, and then use computer processing to achieve some purpose of image

modification. The rapid advancement of computing and mathematics has led to improvements and perfections in digital image processing technology. Image quality enhancement, picture analysis, and image reconstruction are its three components. Because it can only handle linear processes, optical image processing can only be applied in a limited number of scenarios [1].

Digital image processing is also capable of realizing nonlinear processing, allowing for the representation of all operations on mathematical formulas as well as logical linkages on image processing procedures. Now, higher-level image processing can be realized by artificial intelligence algorithms such as neural networks. Digital image processing is mainly composed of the following four aspects. First, image digitization, which is the basic step before the computer processes the image, is to transform the real image into a storage format that the computer can accept.

The image digitization process is divided into two steps: sampling and quantization. Sampling refers to the process of changing continuous quantities in the time domain or space domain into discrete quantities, while quantization refers to the process of approximating the continuous values of signals to a limited number of discrete values. Second, the correlation of digital images is one of the two fundamental characteristics of digital image compression. In other words, there is frequently a significant relationship between adjacent pixels and their corresponding pixels in adjacent frames. Eliminating or reducing these correlations will also cause the reducing of the redundancy in the image information, which means it realizes the compression of digital images [7].

The other one is human visual psychological features. Human vision is insensitive to weak in color resolution and abrupt changes in edges. Using these properties can properly reduce the accuracy of coding in the relative parts, which cause that people do not experience the decline of image quality visually, so as to realize the purpose of digital image compression. Third is image enhancement, which uses different kinds of mathematical means and transformation means to improve the contrast ratio and clarity of the image, so that, for human visual characteristics or machine recognition systems, the processed image is more suitable by using this method. Fourth, image segmentation is to separate the image into multiple specific regions with unique properties, each of which is a continuous set of pixels. In the development of digital image processing, there are many factors that restrict its development. For example, digital image processing has a large amount of information,

occupies the computing speed of the computer, and has certain requirements for storage capacity. At the same time, it is difficult to process the frequency band. Because the development of digital image processing has the characteristics of high-frequency band images, it is particularly critical in frequency band processing, storage and display. Secondly, because the image is the two-dimensional projection of the three-dimensional object, the image in two-dimension can't reconstruct all of the stereoscopic information of the object, so it makes image recognition more difficult in the two-dimensional stage.

In addition, of course, there are many problems in digital image processing that need to be solved by researchers. The root cause of these problems is that it is often not people who process and analyze images, but machines or computers. This non-subjective way of thinking through code makes computers often do not have the ability to judge images through human perception, which is also one of the problems facing the field of digital image processing and artificial intelligence.

Through the research of papers in recent years, it is not difficult to find that digital image processing is mainly used in remote sensing technology, medical observation, human feature recognition and intelligent transportation. By researching these relative studies, it is practical to further sort out the basic context of the development of digital image processing.

2.3.1 RGB Images

RGB (or true colour) images are 3-D arrays that we may consider conceptually as three distinct 2-D planes, one corresponding to each of the three Red (R), Green (G) and Blue (B) colour channels. RGB is the most common colour space used for digital image representation as it conveniently corresponds to the three primary colours which are mixed for display on a monitor or similar device.

In an RGB image, each pixel (the smallest unit of the image) is made up of three color components: one for red, one for green, and one for blue. Each of these components has a value that ranges from 0 to 255, where 0 means no intensity (the color is off) and 255 means full intensity (the color is at its brightest). By adjusting the intensity of these three colors, millions of different colors can be represented.

For example, if a pixel has the values (255, 0, 0), it will appear as pure red. If the values are (0, 255, 0), the pixel will be pure green, and if they are (0, 0, 255), it will

be pure blue. When all three colors are at their maximum intensity (255, 255, 255), the pixel will appear white, and when all are at zero intensity (0, 0, 0), it will appear black.

RGB images are commonly used in digital cameras, computer graphics, and displays because they closely mimic how human eyes perceive color. They are the standard format for most digital images and are widely used in everything from photography to web design and video production.

We can easily separate and view the red, green and blue components of a true-colour image, as shown in Figure 2.5. It is important to note that the colours typically present in a real image are nearly always a blend of colour components from all three channels. A common misconception is that, for example, items that are perceived as blue will only appear in the blue channel and so forth. Whilst items perceived as blue will certainly appear brightest in the blue channel (i.e. they will contain more blue light than the other colours) they will also have milder components of red and green.

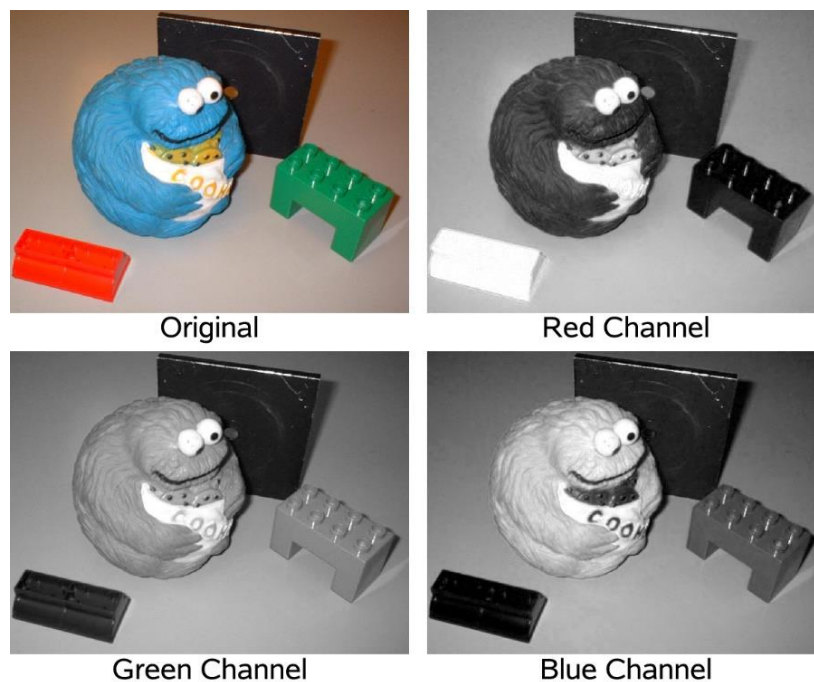


Figure 2.5 Colour RGB Image Separated into Its R, G, B Colour Channels

If we consider all the colours that can be represented within the RGB representation, then we appreciate that the RGB colour space is essentially a 3-D colour space (cube) with axes R, G and B is shown in Figure 2.6.

Each axis has the same range 0 -1 (this is scaled to 0–255 for the common 1 byte per colour channel, 24-bit image representation). The colour black occupies the origin of the cube (position $(0; 0; 0)$), corresponding to the absence of all three colours;

white occupies the opposite corner (position $\delta 1; 1; 1$), indicating the maximum amount of all three colours. All other colours in the spectrum lie within this cube.

The RGB colour space is based upon the portion of the electromagnetic spectrum visible to humans (i.e. the continuous range of wavelengths in the approximate range 400–700 nm). The human eye has three different types of colour receptor over which it has limited (and nonuniform) absorptency for each of the red, green and blue wavelengths. This is why, as we will see later, the colour to grey-scale transform uses a nonlinear combination of the RGB channels.

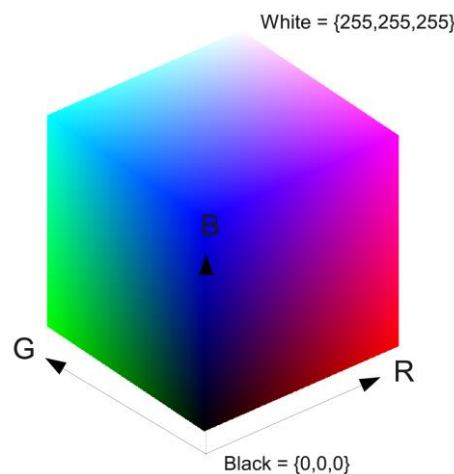


Figure 2.6 An Illustration of RGB Colour Space as 3-D Cube

In digital image processing we use a simplified RGB colour model (based on the CIE colour standard of 1931) that is optimized and standardized towards graphical displays. However, the primary problem with RGB is that it is perceptually nonlinear. By this we mean that moving in a given direction in the RGB colour cube shown in Figure 2.6 does not necessarily produce a colour that is perceptually consistent with the change in each of the channels.

For example, starting at white and subtracting the blue component produces yellow; similarly, starting at red and adding the blue component produces pink. For this reason, RGB space is inherently difficult for humans to work with and reason about because it is not related to the natural way, we perceive colour. As an alternative we may use perceptual colour representations such as HSV.

2.3.2 Color Image to Grayscale Image Conversion

Conversion of a color image to a grayscale image requires more knowledge about the color image. A pixel color in an image is a combination of three colors Red,

Green, and Blue (RGB). The RGB color values are represented in three dimensions XYZ, illustrated by the attributes of lightness, chroma, and hue. Quality of a color image depends on the color represented by the number of bits the digital device could support. The basic color image represented by 8 bits, the high color image represented using 16 bits, the true color image represented by 24 bits, and the deep color image is represented by 32 bits. The number of bits decides the maximum number of different colors supported by the digital device. If each Red, Green, and Blue occupies 8 bits then the combination of RGB occupies 24 bit and supports 16,777,216 different colors. The 24 bit represents the color of a pixel in the color image. The grayscale image has represented by luminance using 8 bits value. The luminance of a pixel value of a grayscale image ranges from 0 to 255. The conversion of a color image into a grayscale image is converting the RGB values (24 bit) into grayscale value (8 bit) [5].

We can convert from an RGB colour space to a grey-scale image using a simple transform. Grey-scale conversion is the initial step in many image analysis algorithms, as it essentially simplifies (i.e. reduces) the amount of information in the image. Although a grey-scale image contains less information than a colour image, the majority of important, feature related information is maintained, such as edges, regions, blobs, junctions and so on. Feature detection and processing algorithms then typically operate on the converted grey-scale version of the image. As we can see from Figure 2.7, it is still possible to distinguish between the red and green apples in grey-scale.



Figure 2.7 RGB Colour Image (Left) to Grey-Scale Image (Right) Conversion

An RGB colour image, I_{colour} , is converted to grey scale, $I_{\text{grey-scale}}$, using the following transformation:

$$I_{\text{grey-scale}}(n,m) = \alpha I_{\text{colour}}(n,m,r) + \beta I_{\text{colour}}(n,m,g) + \gamma I_{\text{colour}}(n,m,b) \quad (2.1)$$

where (n, m) indexes an individual pixel within the grey-scale image and (n, m, r) the individual channel at pixel location (n, m) the colour image for channel c in the

red r , blue b and green g image channels. The grey-scale image is essentially a weighted sum of the red, green and blue colour channels. The weighting coefficients (a , b and g) are set in proportion to the perceptual response of the human eye to each of the red, green and blue colour channels and a standardized weighting ensures uniformity (NTSC television standard, α 0.2989, β 0.5870 and γ 0.1140). The human eye is naturally more sensitive to red and green light; hence, these colours are given higher weightings to ensure that the relative intensity balance in the resulting grey-scale image is similar to that of the RGB colour image.

RGB to grey-scale conversion is a noninvertible image transform: the true colour information that is lost in the conversion cannot be readily recovered.

2.3.3 Perceptual Colour Space

Perceptual colour space is an alternative way of representing true colour images in a manner that is more natural to the human perception and understanding of colour than the RGB representation. Many alternative colour representations exist, but here we concentrate on the Hue, Saturation and Value (HSV) colour space popular in image analysis applications.

Changes within this colour space follow a perceptually acceptable colour gradient. From an image analysis perspective, it allows the separation of colour from lighting to a greater degree. An RGB image can be transformed into an HSV colour space representation as shown in Figure 2.8. Each of these three parameters can be interpreted as follows:

- H (hue) is the dominant wavelength of the colour, e.g., red, blue, green.
- S (saturation) is the ‘purity’ of colour (in the sense of the amount of white light mixed with it).
- V (value) is the brightness of the colour (also known as luminance).

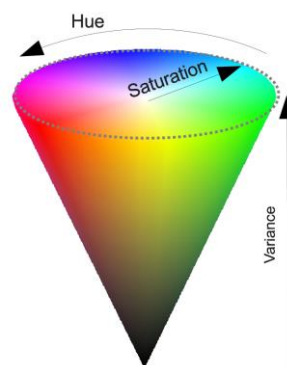


Figure 2.8 HSV Colour Space as 3-D cone

The HSV representation of a 2-D image is also as a 3-D array comprising three channels (h, s, v) and each pixel location within the image, (n, m), contains an (h, s, v) triplet that can be transformed back into RGB for true-colour display. In the MATLAB HSV implementation each of h, s and v are bounded within the range $0 \rightarrow 1$. For example, a blue hue (top of cone, Figure 1.9) may have a value of $h = 0.9$, a saturation of $s = 0.5$ and a value $v = 1$ making it a vibrant, bright sky-blue.

By examining the individual colour channels of images in the HSV space, we can see that image objects are more consistently contained in the resulting hue field than in the channels of the RGB representation, despite the presence of varying lighting conditions over the scene and is shown in Figure 2.9. As a result, HSV space is commonly used for colour-based image segmentation using a technique known as colour slicing. A portion of the hue colour wheel (a slice of the cone, Figure 2.8) is isolated as the colour range of interest, allowing objects within that colour range to be identified within the image. This ease of colour selection in HSV colour space also results in its widespread use as the preferred method of colour selection in computer graphical interfaces and as a method of adding false colour to images.

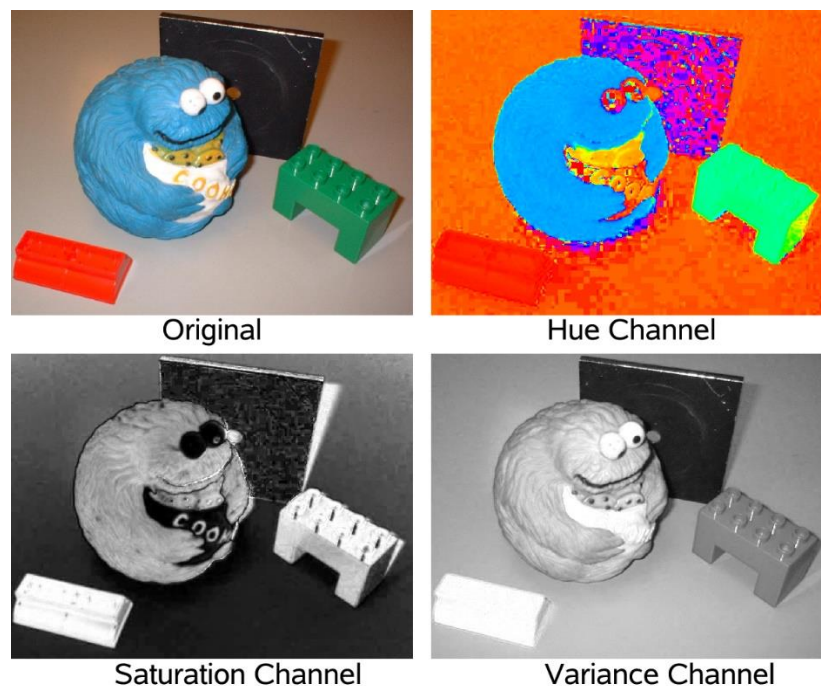


Figure 2.9 Image Transformed and Displayed in HSV Colour Space

2.4 Image Feature Extraction

Feature Extraction uses an object-based approach to classify imagery, where an object (also called segment) is a group of pixels with similar spectral, spatial, and/or

texture attributes. Traditional classification methods are pixel-based, meaning that spectral information in each pixel is used to classify imagery. With high-resolution panchromatic or multispectral imagery, an object-based method offers more flexibility in the types of features to extract.

Feature extraction is a part of the dimensionality reduction process, in which, an initial set of the raw data is divided and reduced to more manageable groups. So, when you want to process, it will be easier. The most important characteristic of these large data sets is that they have a large number of variables. These variables require a lot of computing resources to process. So, Feature extraction helps to get the best feature from those big data sets by selecting and combining variables into features, thus, effectively reducing the amount of data. These features are easy to process, but still able to describe the actual data set with accuracy and originality.

The technique of extracting the features is useful when you have a large data set and need to reduce the number of resources without losing any important or relevant information. Feature extraction helps to reduce the amount of redundant data from the data set. In the end, the reduction of the data helps to build the model with less machine effort and also increases the speed of learning and generalization steps in the machine learning process.

2.4.1 SIFT Algorithm

The Scale-Invariant Feature Transform (SIFT) algorithm is a computer vision technique used for feature detection and description. It detects distinctive key points or features in an image that are robust to changes in scale, rotation, and affine transformations [4].

SIFT works by identifying key points based on their local intensity extrema and computing descriptors that capture the local image information around those key points. These descriptors can then be used for tasks like image matching, object recognition, and image retrieval [8].

SIFT algorithm helps locate the local features in an image, commonly known as the ‘keypoints’ of the image. These keypoints are scale & rotation invariants that can be used for various computer vision applications, like image matching, object detection, scene detection, etc. SIFT algorithm is effective for finding local features of image. The feature points, scales and direction descriptors are used to extract relevant features and

match images. SIFT algorithm has a good effect in feature extraction and detection. Changing the rotation angle and brightness does not affect the scale.

The Scale Space was constructed to detect the extreme point and obtain the scale invariance. Since Koendetink found that there was only one way to scale transformation, and that was to use the Gaussian kernel, image features were extracted at different scales to achieve scale invariance. First build the Gaussian Pyramid. Then the Dog pyramid was built. Finally, extremum detection is carried out on the basis of DOG pyramid [6].

i. Building the Gaussian pyramid. According to the two-dimensional Gaussian filtering function, Gaussian filtering is performed on the image.

$$G(x, y, \sigma) = \frac{1}{2\pi\sigma^2} e^{-(x^2+y^2)/2\sigma^2} \quad (2.2)$$

Where σ is called variance. The representation (x, y) in different scale Spaces can be obtained by the Gaussian kernel convolution.

$$L(x, y, \sigma) = G(x, y, \sigma) \times I(x, y) \quad (2.3)$$

σ is the scale space factor.

If it gets lower, the image gets smoother. Large scale is a general feature of the image. Small scale is an image detail feature. Therefore, the size of the scale factor is very important for the scale space description.

The Gaussian pyramid is built according to the scale function. The first layer of the first order of the Gaussian pyramid is the original image. The Gaussian pyramid has o and s levels. The ratio of two adjacent levels in the same order is k. For example, if the ratio between level 1 and level 2 is σ , then the ratio between level 1 and level 3 is $k\sigma$. And each order image is one half the size of the previous order image.

ii. Build the Dog Gauss pyramid based on the Gauss pyramid. A layer of Dog Gaussian pyramid can be obtained by the difference of two adjacent scale space functions of the same order. The expression for Dog is defined as (2.4)

$$D(x, y, \sigma) = (G(x, y, k\sigma) - G(x, y, \sigma)) \times I(x, y) = L(x, y, k\sigma) - L(x, y, \sigma) \quad (2.4)$$

iii. The extremum points are detected and the key points are obtained. The 26 pixel values adjacent to each pixel in the Dog image need to be compared. The adjacent pixel points need to be taken from the middle layer on the same order (except for the bottom and top layer on the same order). Through comparison, the maximum or

minimum value of the point is needed for the selected point, and then the key points are determined by eliminating the selected points and edge points with low contrast and noise sensitivity.

And then, the system determines the direction value of key points. The SIFT algorithm reduces computational requirements to the extent that a large number of feature description points are retained in the image by using the pyramid image structure. The formula is as (2.5) and 2.6).

$$m(x, y) = \sqrt{(L(x+1, y) - L(x-1, y))^2 + (L(x, y+1) - L(x, y-1))^2} \quad (2.5)$$

$$\theta(x, y) = \tan^{-1}((L(x, y+1) - L(x, y-1)) / (L(x+1, y) - L(x-1, y))) \quad (2.6)$$

The adjacent pixel values determine the gradient direction of the key points. L is the spatial scale function of the key point, and (x, y) is its adjacent pixel region of the key point. Gradient histogram is used to calculate the gradient direction of the adjacent pixel region. The horizontal axis of the gradient histogram represents the value of the gradient direction of the adjacent pixel region. The magnitude of the vertical axis is the neighborhood pixel gradient value. The horizontal axis of the gradient histogram ranges from 0° to 360° . Every 10° is one unit. There are 36 units. The main peak of the histogram of the gradient direction represents the main direction of the key point. If there are other peaks equal to 80% of the main peak value, the secondary direction of the key point.

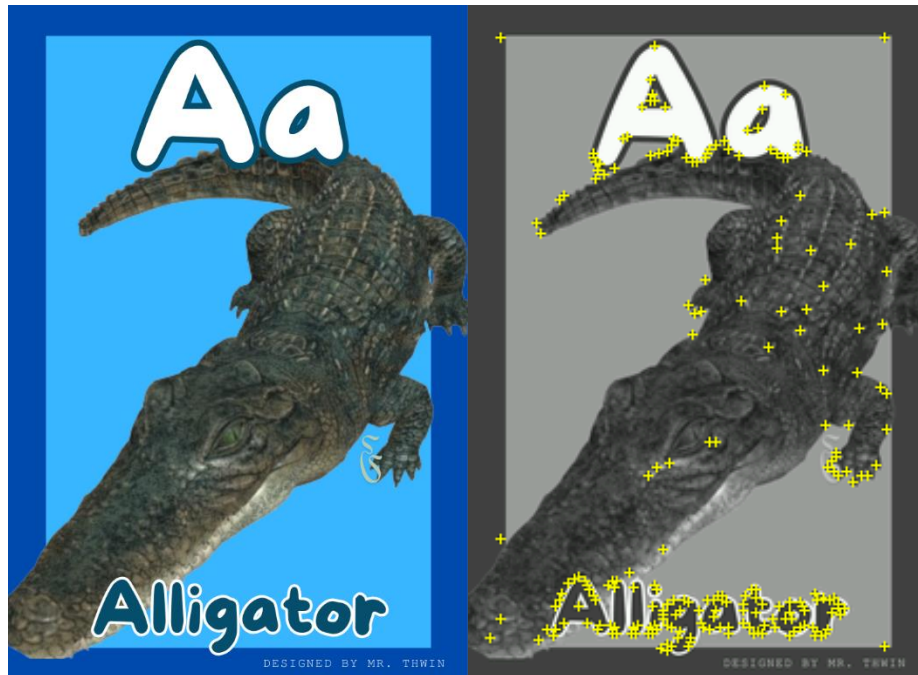


Figure 2.10 Alphabet “A” Flashcard’s Feature Extraction Using SIFT

The source code for implementing SIFT algorithm in C# programming language is described in Chapter 3, System Implementation article.

2.5 Augmented Reality App Development with Vuforia Engine

Vuforia Engine is the most widely used platform for AR development, with support for the majority of phones, tablets, and eyewear. Developers can easily add advanced computer vision functionality to Android, iOS, and UWP apps, to create AR experiences that realistically interact with objects and the environment.

Vuforia Engine is a powerful tool for developing Augmented Reality (AR) applications. It enables developers to create AR experiences by recognizing and tracking images, objects, and environments in real time. Vuforia uses advanced algorithm of SIFT for feature extraction and matching, allowing digital content such as 3D models and videos to be overlaid onto the physical world. It integrates seamlessly with platforms like Unity, making it easier to build interactive and immersive AR applications across various devices. Vuforia's flexibility and robust features make it a popular choice for AR development.

2.5.1 Vuforia Target Manager Development

The Vuforia Target Manager allows you to create, manage, and download Device Databases for image-based Vuforia Targets. Device Databases provide your AR application with a locally accessible Vuforia target. A Device database – also referred to as datasets in the SDK – is a set of runtime assets and authoring files for a Vuforia target. The database is locally accessible and contains the targets that your application is tracking. Download the databases from the Target Manager or get them from one of the Vuforia tools capable of generating databases.

The Vuforia Target Manager Database is a key component in developing Augmented Reality (AR) applications using the Vuforia Engine. This database stores the visual targets that the AR application will recognize and interact with, such as images, objects, or surfaces.

In the development process, images are uploaded to the Vuforia Target Manager, where they are analyzed and processed to extract key features using algorithms like SIFT (Scale-Invariant Feature Transform). These features are then stored in the database as "targets." Each target is associated with specific digital content,

such as 3D models or videos, that will be rendered when the target is recognized by the application.

When users scan a physical image or object with their device, the Vuforia Engine compares the camera input with the stored targets in the Vuforia Target Manager Database. If a match is found, the corresponding digital content is displayed in the AR environment.

The screenshot shows the Vuforia Engine developer portal's Target Manager interface. The top navigation bar includes links for Home, Downloads, Library, Support, and Pricing, along with a user account section (My Account | Log Out). The Target Manager section is highlighted in green. Below the navigation bar, there's a 'Generate Database' button. A search bar is present, followed by a table listing three databases. The table has columns for Database, Type, Targets, and Date Modified. Each row also includes a trash icon for deletion. At the bottom, there's a pagination bar showing 'Showing 1-3 of 3' and a 'Refresh' button.

Database	Type	Targets	Date Modified
3D_AR	Device	26	Aug 24, 2024
AR_Mag	Device	5	Aug 24, 2024
Eng_Poem	Device	4	Aug 24, 2024

Figure 2.11 Building Three Databases in Target Manager Database

2.5.2 Using Image Target

Image Targets represent images that Vuforia Engine can detect and track. The image is tracked by comparing extracted natural features from the camera image against a known target resource database. Once the Image Target is detected, Vuforia Engine will track the image and augment your content seamlessly using the best image-tracking technology in the market.

Image Targets can be created using JPG or PNG images in RGB or grayscale with the Target Manager. The size of the input images must be 2.25 MB or less and have a minimum width of 320 pixels. Features extracted from these images are stored in a cloud or device database, of which the latter can be downloaded and packaged together with the application. The database can then be used by Vuforia Engine for runtime comparisons.

Using Image Targets in Vuforia is a powerful way to create immersive AR experiences, as it allows developers to anchor virtual objects to real-world images,

making them interactable and visually engaging for users. This technique is widely used in educational apps, marketing campaigns, games, and many other fields.

2.5.3 Designing and Developing Image-Base Targets

When it comes to image-based targets, there are a range of factors that define how well it tracks and its target star rating when uploaded to the Target Manager. This guide will provide you with insight into what makes a good Image Target.

Image-based targets come in various forms: simple, flat Image Targets, curled targets in form of cylindrical shapes, or Multi-Targets in the composition of a box. They all share similar requirements for having rich detail and a good star rating. This article focuses on Image Targets, but the same factors apply to Vuforia's other image-based targets.

2.5.3.1 Attributes of an ideal image target

Image Targets that possess the following attributes will enable the best detection and tracking performance from the Vuforia Engine.

Table 2.2 Attributes of an Ideal Image Target

Attribute	Example
Rich in detail	Street scene, group of people, collages and mixtures of items, and sport scenes are good examples.
Good Contrast	Images with bright and dark regions and well-lit areas work well.
No repetitive patterns	Employ unique features and distinct graphics covering as much of the target as possible to avoid symmetry, repeated patterns, and feature-less areas.
Format	Must be 8- or 24-bit PNG and JPG formats; less than 2 MB in size; JPGs must be RGB or greyscale (no CMYK).

Figure 2.12 show Image Target with coordinate axes for explanation. This image is fed into the online [Target Manager](#) to create the target database.



Figure 2.12 Input Image Target -Feature Extract

2.5.3.2 Target star rating

Image Targets are detected based on natural features that are extracted from the target image and then compared at run time with features in the live camera image. The star rating of a target ranges between 1 and 5 stars. Although, targets with low rating (1 or 2 stars) can usually detect and track, aim for targets with 4 or 5 stars for best results.

2.5.3.3 Natural features and image ratings

An augmentable rating defines how well an image can be detected and tracked using the Vuforia Engine. This rating is displayed in the Target Manager and is returned for each uploaded target. The augmentable rating can range from 0 to 5 for any given image. The higher the augmentable rating of an image target, the stronger the detection and tracking ability it contains. A rating of zero indicates that a target is not tracked at all by the AR system, whereas a star rating of 5 indicates that an image is easily tracked by the AR system.

(i) Rating – Features

A feature is a sharp, spiked, chiseled detail in the image, such as the ones present in textured objects. The image analyzer represents features as small yellow crosses. Increase the number of these details in your image and verify that the details create a non-repeating pattern. Table 2.3 demonstrates how to analyze features in an input image in Vuforia Engine.

Table 2.3 Analyzing Features in an Input Image

Shapes	Properties
	A square contains four features for each one of its corners.
	A circle contains no features as it contains no sharp or chiseled detail.
	This object contains only two features for each sharp corner. Note: According to the definition of a feature, soft corners and organic edges are not marked as features.

(ii) Comparison of Two Images

Inspect the two images and notice the lack of feature points in the first image of Figure 2.13. It is always advisable to design and find images that show a significant number of points. As described later, an even distribution of feature points also improves tracking and the robustness of your augmentations: your augmented content is steadier when they are placed on top of a feature rich area of your Image Target.



Figure 2.13 Comparison of Low-Star Rating and High-Star Rating

2.5.3.4 The feature-exclusion buffer

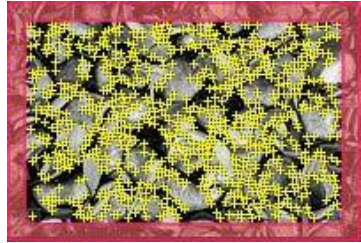
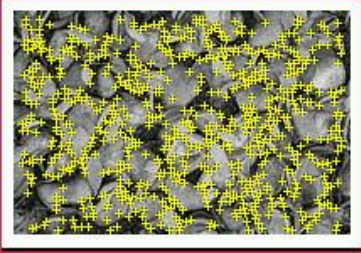
A feature's exclusion buffer surrounds the inset of an uploaded image. This buffer area is about 8% wide and it does not pick up any features, even if features do exist within that zone. The first row of the following table shows where the shaded area in red does not contain any features, even though visible features are present in this zone. By using a Feature-Exclusion Buffer, systems can reduce processing time and avoid potential errors caused by extraneous data. This leads to more efficient and reliable performance, especially in complex tasks like AR, where quick and accurate feature matching is essential.

When an image is analyzed, numerous features such as edges, corners, and textures are detected. However, not all of these features are useful for the task at hand. For example, in an AR application, some background details in an image might not contribute to the recognition of a specific object or target. The Feature-Exclusion Buffer

helps to exclude these irrelevant features, so the system can focus on what really matters. This buffer works by temporarily holding features that are identified during the initial image analysis and then applying a filtering process to exclude or ignore features that don't meet certain criteria. These criteria could be based on factors like the size, shape, or orientation of the feature, or its relevance to the specific task.

You can avoid this feature-exclusion buffer situation by adding a white 8% buffer around the image stored in Target Manager, as shown in the lower row of the following table 2.4.

Table 2.4 The Feature-Exclusion Buffer

	Uploaded Image	Analyzed Image (Red Marking)
Original Image Target		
Image Target with border		

But consider that those features will be helpful only when it can be guaranteed that during the run time execution the target will lie on a surface with a unique color that does not itself have features.

2.6 3D Modeling

Three-dimensional (3D) modeling is a comprehensive process that begins with data collection and ends with a visually interactive 3D model on a computer. Often, people think of 3D modeling as just converting a set of measured points into a mesh or textured surface, but it's actually a broader process that involves reconstructing the entire object [2].

3D modeling has been a significant research area in graphics, vision, and photogrammetry communities for many years. It is essential in various applications,

such as inspection, navigation, object identification, visualization, and animation. Recently, 3D modeling has become particularly important for cultural heritage digital archiving. This is due to its use in documenting and preserving cultural artifacts, enabling virtual tourism, providing educational resources, and allowing interaction with valuable objects without risking damage.

Creating a 3D model that meets the high demands of accuracy, photo-realism, and detail is challenging. It requires selecting the right modeling technique, which can vary depending on the application. Digital models are becoming more common and accessible on the internet and can be viewed on low-cost computers. However, making a precise and photo-realistic model of a complex object still requires significant effort.

3D object measurement and reconstruction techniques are generally classified into two types: contact methods and non-contact methods. Contact methods involve physical tools like measuring machines or rulers, while non-contact methods use technology such as X-rays, photogrammetry, or laser scanning.

This article focuses on modeling from reality, which involves capturing real-world objects using sensors, rather than creating artificial models with software like 3DMax or Maya. The most common approach today involves non-contact systems that use light waves, especially sensors. These sensors can be active, directly providing the necessary 3D coordinates, or passive, which require additional processing to derive the coordinates.

Once the data is collected, it's organized, and a consistent polygonal surface is created to build a realistic 3D model. This model can then be enhanced with textures to achieve a photo-realistic visualization.

CHAPTER 3

SYSTEM DESIGN AND IMPLEMENTATION

This chapter showcases the overview of system design, system flow and details of implementation steps and result of the system.

3.1 System Design

The design of the system is shown in Figure 3.1.

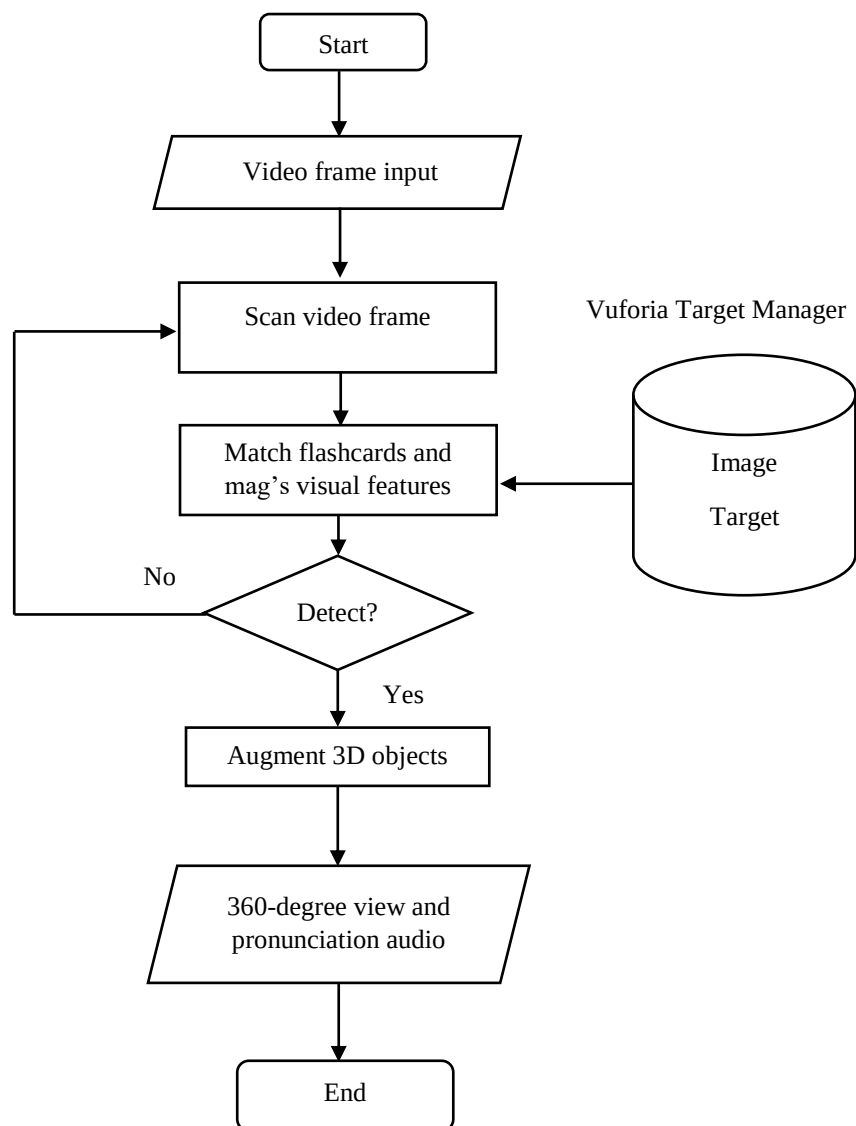


Figure 3.1 System Design

Users begin by opening the application on their mobile devices and then scan either flashcards or magazine pages with their device's camera. When a user scans an

image, the system takes the camera input and compares it to the preprocessed images stored in the Vuforia Image Target Database. This comparison is done using a technique called SIFT (Scale-Invariant Feature Transform), which extracts and matches features from the scanned image to those in the database. If the system finds a match, the corresponding 3D model or live-motion video player is displayed on the screen, appearing as though it is coming out of the flashcard or magazine page. Users can interact with these 3D models by rotating them in a full 360-degree view, making the learning experience more engaging. Additionally, the system provides audio for pronunciation and videos related to the content.

3.2 System Flow

According to the system flow shown in Figure 3.2, the process begins with collecting images of flashcards and magazine pages. These images are first converted from color (RGB) to grayscale using the RGB-to-Grey scale technique. This conversion makes it easier to focus on the important features of the images. Next, the system uses the SIFT (Scale-Invariant Feature Transform) algorithm to extract key features from these grayscale images. These extracted features are then stored in the Vuforia Target Manager database, where they are ready for matching.

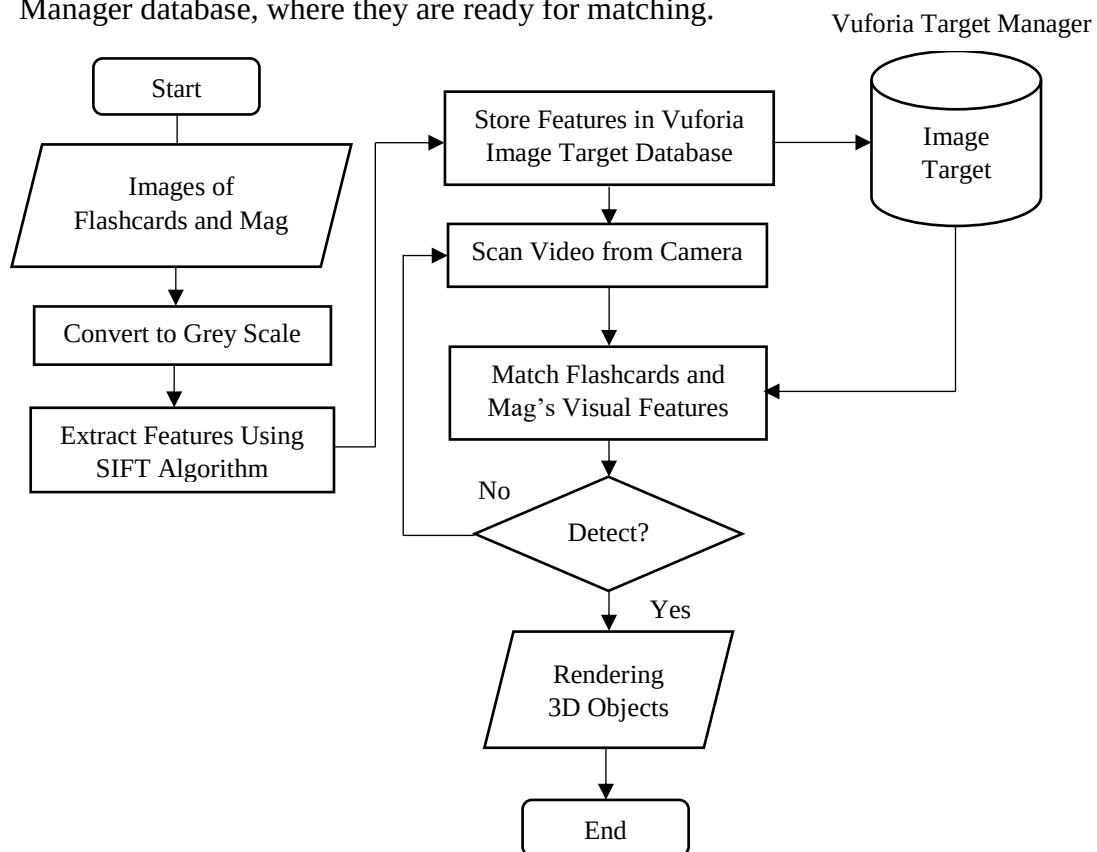


Figure 3.2 System Flow

When users open the application and scan a flashcard or magazine page with their mobile device, the system captures the camera input and compares it to the stored features in the Vuforia database. This comparison, called SIFT Feature Matching, checks if the scanned image matches any of the preprocessed images. If a match is found, the system retrieves the corresponding 3D model or live-motion video from the Unity application database. This content is then displayed on the screen, bringing the image to life. Once the process ends, complete the user’s interactive experience.

3.3 System Implementation

In this article, the step-by-step implementation of the system is described completely. Firstly, the developer needs to download Unity Software Editor. The Unity Editor is a popular and useful authoring platform for creating cutting-edge augmented reality experiences for handheld devices and digital eyewear. Visit <https://unity.com> to download and set up Unity. Add the Vuforia Engine Extension to the project, or start with [Core Samples](#) in the Unity Asset Store.

In Unity Hub, a new project named “AR” was created by selecting the 3D core template.

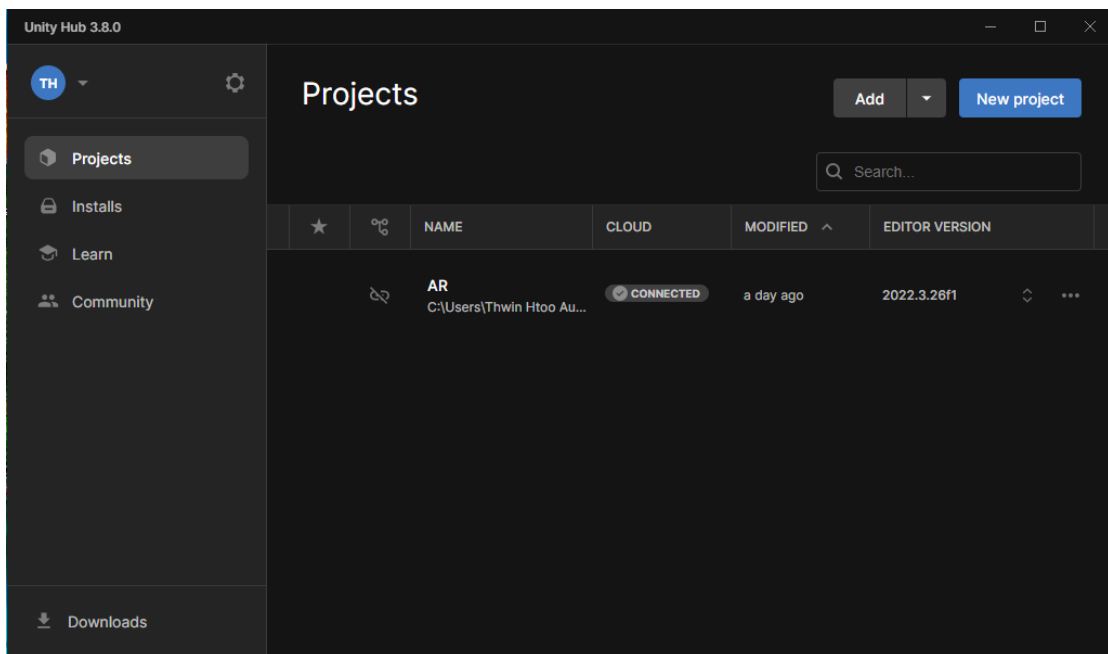


Figure 3.3 AR Project Initiation in Unity Editor

Download the extension with the installer script from the Developer portal: [Vuforia Engine Extension for Unity](#). Import the downloaded Unity Asset Package from the menu Assets -> Import Package -> Custom Package.

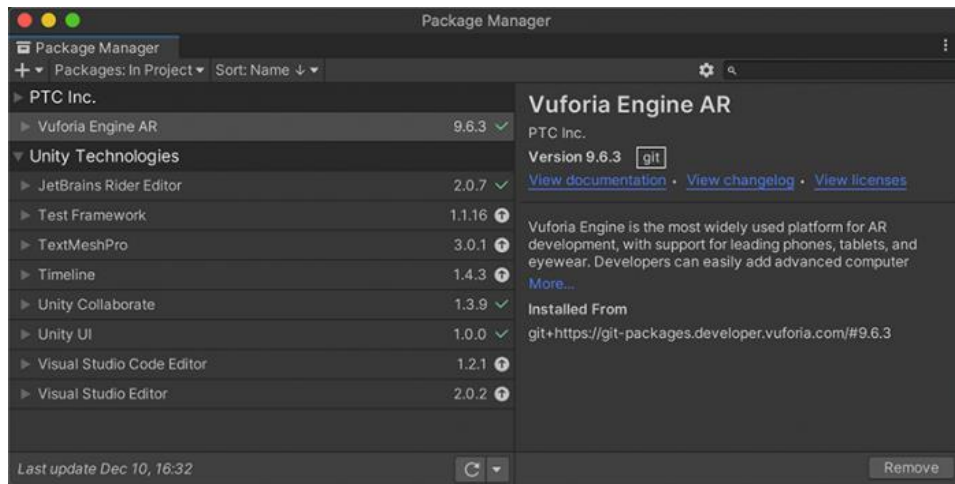


Figure 3.4 Importing Vuforia Engine Packet in Unit

At <https://developer.vuforia.com/auth/login>, create Vuforia developer account to download software, get license keys and participate in the Vuforia Community.

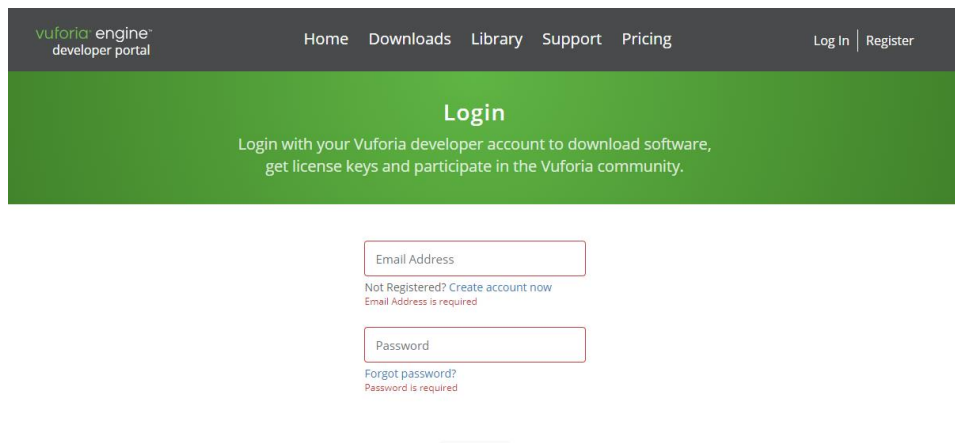


Figure 3.5 Creating Vuforia Developer Account

Three databases named 3D AR, AR Mag and Eng. Poem are created in Vuforia Target Manager Database as shown in Figure 3.6.

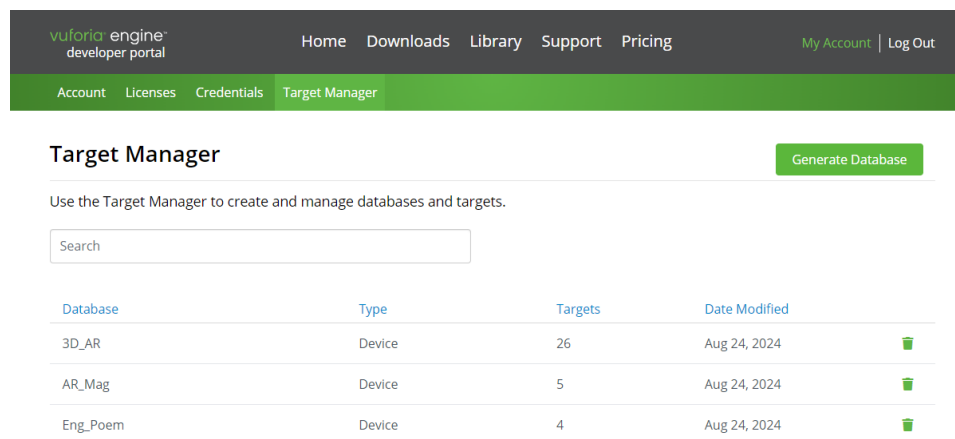


Figure 3.6 Creating three databases named 3D AR, AR Mag and Eng. Poem

vuforia engine[™]

developer portal

Home

Downloads

Library

Support

Pricing

My Account

Log Out

Account

Licenses

Credentials

Target Manager

Target Manager

>

3D_AR

3D_AR

Edit Name

Type: Device

Targets (26)

Add Target

Download Database (All)

☐	Image	Target Name	Type	Rating ①	Status ▾	Date Modified
☐		Iguana	Image	★★★★★	Active	Aug 24, 2024
☐		Queen	Image	★★★★☆	Active	Aug 24, 2024
☐		Whale	Image	★★★☆☆	Active	Aug 21, 2024
☐		Robot	Image	★★★★☆	Active	Aug 21, 2024
☐		Parrot	Image	★★★★☆	Active	Aug 21, 2024
☐		Lung	Image	★★★☆☆	Active	Aug 21, 2024
☐		Heart	Image	★★★★☆	Active	Aug 21, 2024
☐		Girl	Image	★★★★☆	Active	Aug 21, 2024
☐		Fish	Image	★★★★☆	Active	Aug 21, 2024
☐		Earth	Image	★★★★★	Active	Aug 21, 2024
☐		Dog	Image	★★★★☆	Active	Aug 21, 2024
☐		Car	Image	★★★★★	Active	Aug 21, 2024
☐		Boat	Image	★★★★★	Active	Aug 21, 2024
☐		Alligator	Image	★★★★★	Active	Aug 21, 2024
☐		Tree	Image	★★★★★	Active	Aug 13, 2024
☐		Sun	Image	★★★★☆	Active	Aug 13, 2024
☐		Zebra	Image	★★★★★	Active	Aug 13, 2024
☐		Yacht	Image	★★★★★	Active	Aug 13, 2024
☐		Xray	Image	★★★★★	Active	Aug 13, 2024
☐		Vegetable	Image	★★★★★	Active	Aug 13, 2024
☐		University	Image	★★★★★	Active	Aug 13, 2024
☐		Octopus	Image	★★★★★	Active	Aug 13, 2024
☐		Nest	Image	★★★★★	Active	Aug 13, 2024
☐		Monkey	Image	★★★★☆	Active	Aug 13, 2024
☐		Kidney	Image	★★★★★	Active	Aug 13, 2024

Showing 1-25 of 26

<<

<

1

2

>

>>

Last updated: Today 12:10 AM

Refresh

25 per page ▾

Figure 3.7 Building the database of 26 A-Z flashcards

In 3D-AR database, 26 A-Z flashcards images are uploaded as shown in Figure 3.7 . These images are first converted from color (RGB) to grayscale using the RGB-to-Grey scale technique. This conversion makes it easier to focus on the important

features of the images. Next, the system uses the SIFT (Scale-Invariant Feature Transform) algorithm to extract key features from these grayscale images which are later stored in the Vuforia Target Manager database, where they are ready for matching.

Figure 3.7, and 3.8 show the database of 26 A-Z flashcards, and the result of stored image of these A-Z flashcards. These images are first converted from color (RGB) to grayscale using the RGB-to-Grey scale technique. This conversion makes it easier to focus on the important features of the images. Next, the system uses the SIFT (Scale-Invariant Feature Transform) algorithm to extract key features from these grayscale images. These extracted features are then stored in the Vuforia Target Manager database, where they are ready for matching.



Figure 3.8 Feature Extracted ABC Flashcard Images Using SIFT Algorithm

The source code of implementing SIFT algorithm in C# programming language will be available at https://drive.google.com/file/d/1QLC3imlvNYWVVBu6e5h0v0fTOzrB0rta/view?usp=drive_link

The features of other D-Z flashcards image are extracted in the same way and stored in Vuforia Target Manager Database. Figure 3.9 shows the database of AR-Mag. Five magazine papers was uploaded in this database.

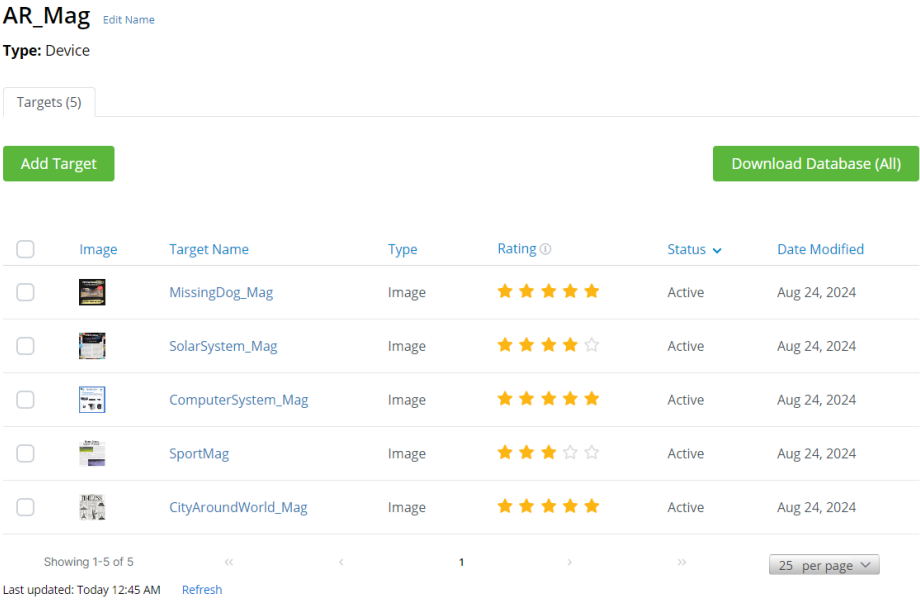


Figure 3.9 Creating AR-Mag Database



Figure 3.10 Cities Around the World Magazine Feature Extraction

For example, Figure 3.10 shows the feature extracted image of Cities Around the World Magazine. The others four magazines are extracted in same way and stored in Vuforia Target Manager Database. Figure 3.11 shows the database of Eng. Poem. Four poem images was uploaded in this database.

Target Manager > Eng_Poem

Eng_Poem

Edit Name

Type: Device

Targets (4)

Add Target

Download Database (All)

☐	Image	Target Name	Type	Rating	Status	Date Modified
☐		TwinkleStar_Poem	Image	★★★★★	Active	Aug 24, 2024
☐		BlackSheep_Poem	Image	★★★★★	Active	Aug 24, 2024
☐		BabyShark_Poem	Image	★★★★★	Active	Aug 24, 2024
☐		ABCD_Poem	Image	★★★★★	Active	Aug 24, 2024

Figure 3.11 Creating Eng. Poem Database



Figure 3.12 Eng. Poem Flashcards Feature Extraction

For example, figure 3.12 shows feature extraction of three English Poem Flashcard images. The other poem flashcard image is extracted in same way and stored in Vuforia Target Manager Database.

And then these database files are imported in Unity Editor as database package file as shown in Figure 3.13 and these database package will be used for further image matching.

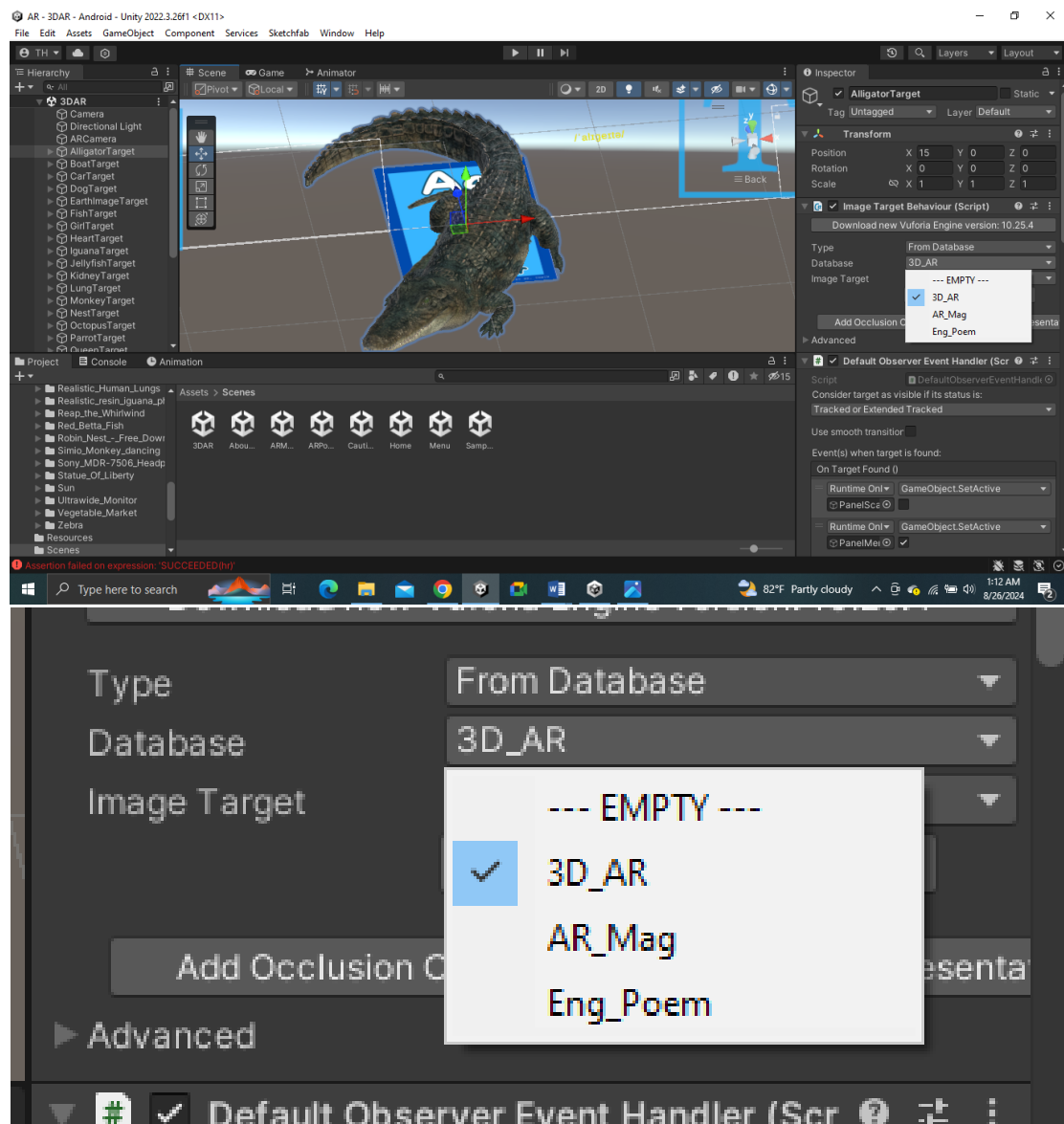


Figure 3.13 Three Database Uploading for Image Target

There will be three databases uploaded called 3D_AR, AR_Mag and Eng_Poem. For the 3D-AR image target feature, there will be 26 image targets, For the AR-Mag image target feature, there will be 5 image targets, and 4 for Eng. Poem feature. The implementation code for image target are written with C# Language. For the control of 3D objects, these scripts codes are written in C# language.

The script code for 3D object rotation is as follows:

```
using System.Collections;

using System.Collections.Generic;

using UnityEngine;

public class rotateObject : MonoBehaviour{

    public GameObject objectRotate;

    public float rotateSpeed = 30f;

    bool rotateStatus = false;

    public void RotateObject()

    {    if(rotateStatus == false) {

            rotateStatus = true; }

        else{ rotateStatus = false; }    }

    void Update()    {

        if (rotateStatus == true){

            objectRotate.transform.Rotate(Vector3.up,rotateSpeed *

Time.deltaTime);

        }

    }

}
```

The script code for 3D object scale up is written as follows:

```
using System.Collections;

using System.Collections.Generic;

using UnityEngine;

public class ScaleIn : MonoBehaviour

{    public GameObject Object;
```

```

    public float Scale = 0.001f;

    public void ScaleInTask()

    {

        Object.transform.localScale += new Vector3(Scale, Scale, Scale);

    }

}

```

The script code for 3D object scale down is written as follows:

```

using System.Collections;

using System.Collections.Generic;

using UnityEngine;

public class scaledown : MonoBehaviour

{
    public GameObject Object;

    public float Scale = 0.001f;

    public void ScaleDownTask()

    {

        Object.transform.localScale -= new Vector3(Scale, Scale, Scale);    } }

```

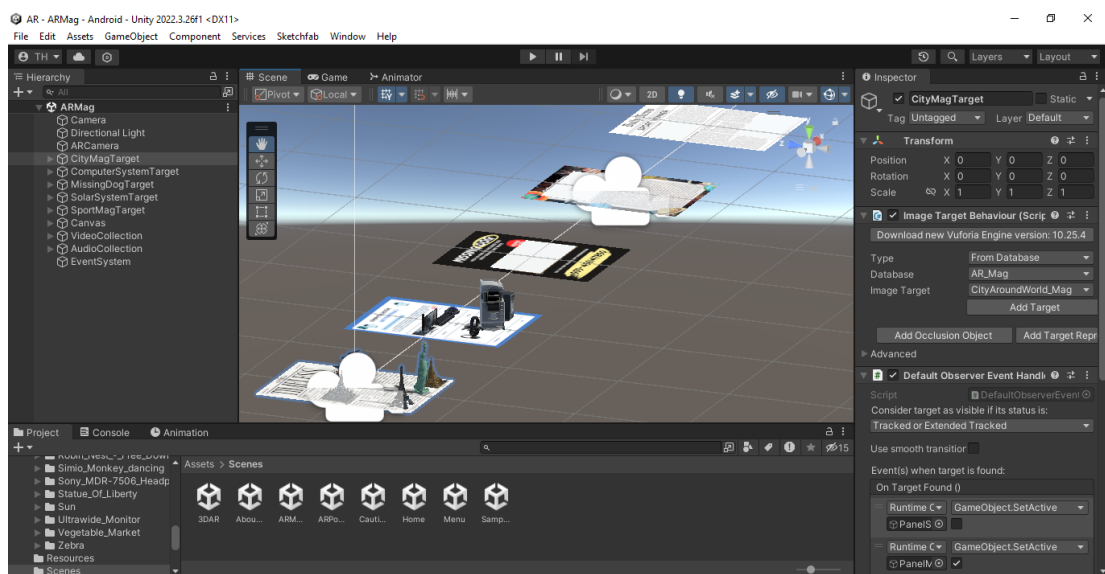


Figure 3.14 AR Mag Feature Implementation

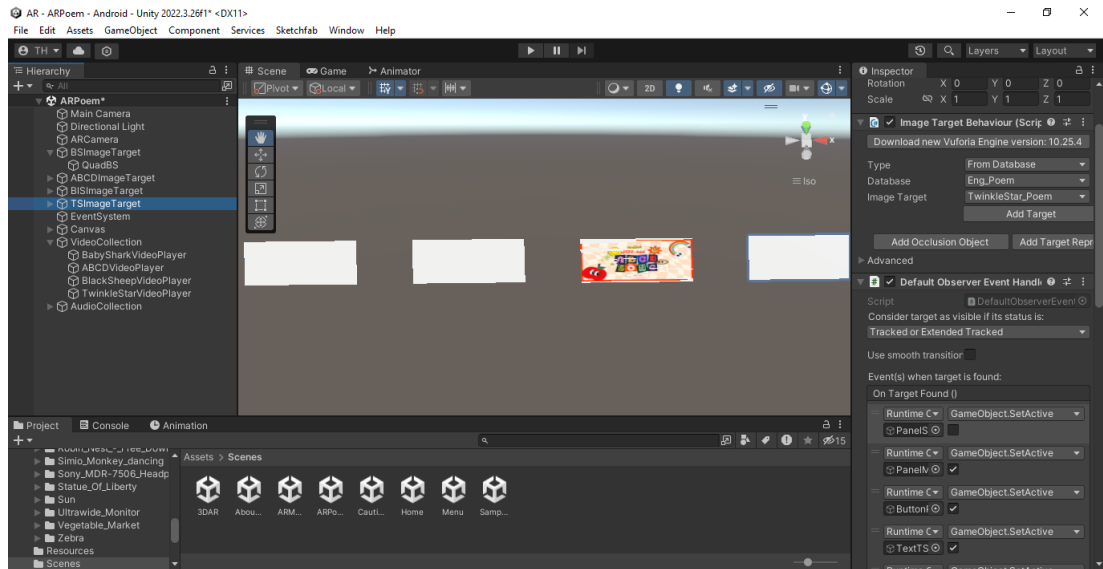


Figure 3.15 Eng. Poem Feature Implementation

Figure 3.14 and 3.15 show the implementation process for AR-Mag and Eng. Poem Feature. This system is about AR application development using SIFT algorithm, and it makes me difficult to discuss each process in application development in detail. So, the source code of my application for studies is uploaded below the link.

https://drive.google.com/drive/folders/1jhin81QUcZhR_vMH14isvSnFd-qONFBt?usp=drive_link

3.4 Output of the System

The output user interface (UI) designs of AR-KID Application are shown in the following figures. The user can download the images of A-Z flashcards, AR Magazine Pages and Eng. Poem Flashcards at the following link.

https://drive.google.com/drive/folders/1MGE50GxQV5XUjdFt509-NAqS5fyylaX8?usp=drive_link

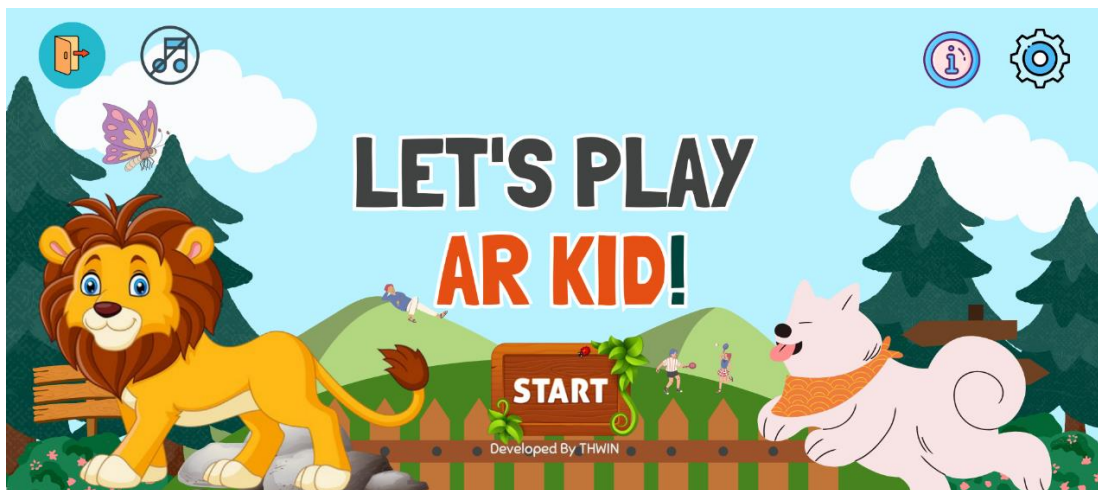


Figure 3.16 AR-KID Application's Home Page UI Design

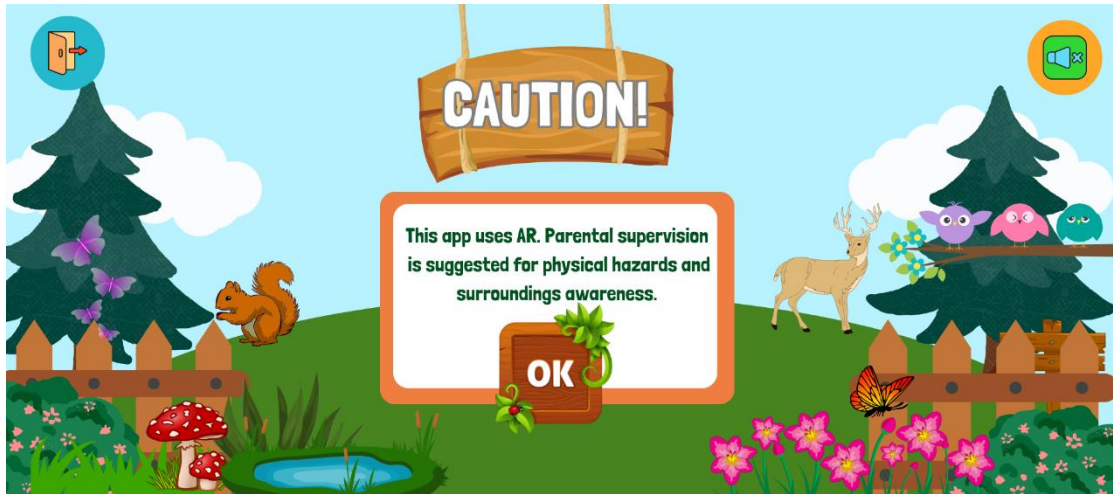


Figure 3.17 AR-KID Application's Caution Page UI Design

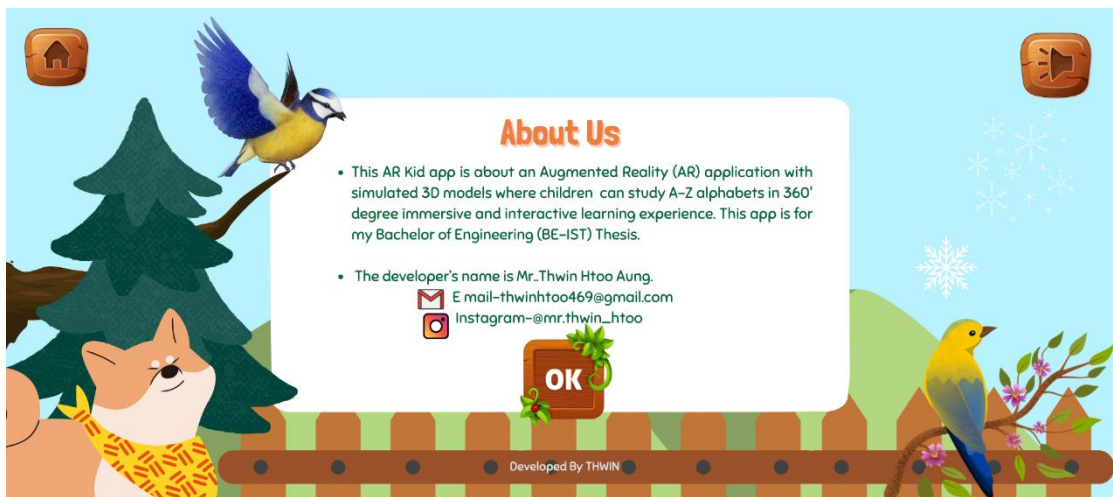


Figure 3.18 AR-KID Application's About Us UI Design

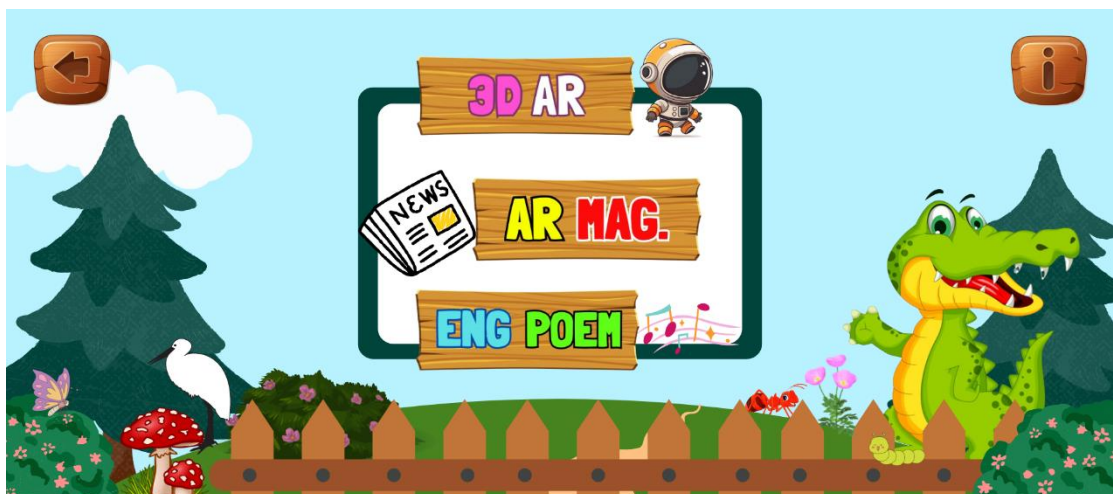


Figure 3.19 AR-KID Application's Menu Page UI Design

The user interface (UI) design of this application is shown in the above Figures 3.16, 3.17, 3.18 and 3.19. When the user clicks 3D AR menu, he can scan in total of 26 flashcards with mobile phone camera.

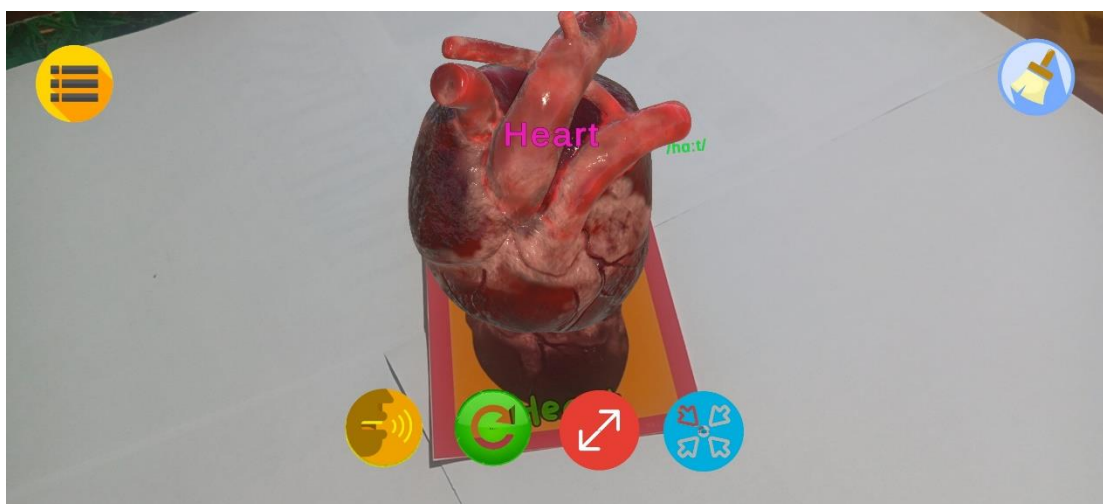


Figure 3.20 Example Scanning of “H” Flashcard in 3D AR Menu

For example, when user scans “H” alphabet flashcard, the output will be as shown in Figure 3.20. There will be four features for each flashcard scanning- vocabulary pronunciation audio, 360-degree rotation of associated 3D model, scaling up the size of this 3D model and scaling down. The others 25 cards will also have their associated 3D models and four control features of these models.

And when the user clicks AR Mag menu, and scan “Cities Around World” magazine paper, the result will be like Figure 3.21. He can scan five magazine papers- “Cities Around World”, “Computer System and Peripherals”, “Solar System”, “Sport Mag” and “Missing Dog”.

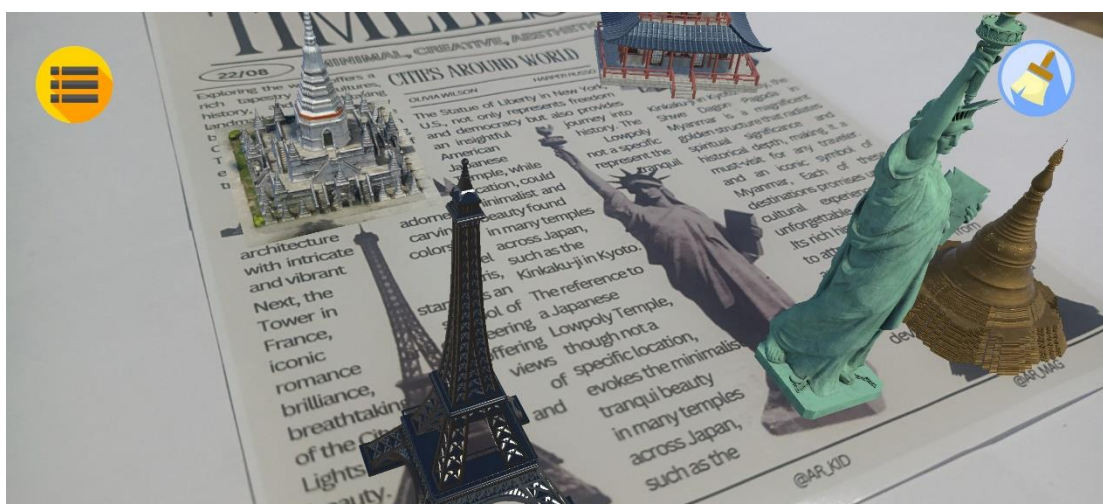


Figure 3.21 Scanning of “Cities Around World” Magazine in AR MAG

The following Figure 3.22 shows the result of scanning “Baa Baa Black Sheep” English poem Flashcard in Eng. Poem menu. In total, the users can scan four English poem flashcards, namely “Baa Baa Black Sheep”, “Twinkle Twinkle Little Star”, “Baby Shark” and “ABCD Poem”.



Figure 3.22 Scanning of Poem Flashcard in Eng. Poem Menu

CHAPTER 4

CONCLUSION

4.1 Conclusion

This chapter wraps up the system by summarizing the key outcomes of the application. It outlines the advantages and limitations of the system, and also explores potential avenues for extending the work and enhancing its current performance. This system set out to address the educational challenges faced by children in Myanmar, particularly in the realm of English vocabulary acquisition, by leveraging Augmented Reality (AR) technology.

The development of the “AR-KID” application demonstrates the potential of AR to transform traditional educational methods into more immersive and engaging learning experiences. By integrating AR with 3D models, interactive content, and multimedia elements, AR-KID successfully bridges the gap between the virtual and real worlds, offering a novel approach to vocabulary learning.

This system has explored the development and implementation of an Augmented Reality (AR) application, AR-KID, designed to enhance the educational experience for children in Myanmar by integrating digital content with traditional learning methods. Through the use of AR technology, the application provides an immersive, interactive platform for learning English vocabulary, making education more engaging and effective.

AR-KID was designed with three primary features which are 3D AR, AR Mag, and Eng. Poem and each targets different aspects of the learning experience. These features allow children to interact with educational content in ways that traditional methods cannot match, making learning more interactive, engaging, and effective. The use of the Unity platform and Vuforia Engine further enhances the application’s capability, enabling seamless image recognition and content delivery, which are crucial for creating a dynamic learning environment.

In conclusion, the “AR-KID” application is not just a tool for learning English vocabulary; it represents a step towards a future where education is more accessible, engaging, and effective for all children. This system underscores the importance of

integrating advanced technologies like AR into educational frameworks to enhance learning outcomes. While AR-KID has shown promising results, further research and development could expand its capabilities, integrating more subjects and features to benefit a wider audience. This would further solidify AR's role as a transformative force in education, particularly in developing regions.

4.2 Benefits of the System

This system aligns with United Nation's SDG No.4 Quality education. This system proposes a solution by using Augmented Reality (AR) technology as a teaching aid to revolutionize traditional teaching methods and upgrade the quality of education for Myanmar children.

This system makes learning more fun and engaging for children. By using 3D models and AR technology, it grabs their attention and helps them learn new words more effectively. Children can interact with the content, which makes learning more interesting and easier to understand. The app also helps them see and understand things better, like when a 3D alligator pops up from a flashcard, making the word more memorable.

AR-KID application can be used on different devices, so it is easy to use in various educational settings. The app can also grow with more content, making it flexible for future needs. The app encourages kids to learn on their own, at their own pace, which builds their confidence. The game-like features make learning feel like play, keeping them motivated and excited to learn. Lastly, this system provides a rich learning experience without needing a lot of physical resources, making it a cost-effective solution for places with fewer educational materials, like Myanmar.

4.3 Limitations of the System

This AR system has only three main features: 3D AR, AR Mag, and Eng. Poem. In 3D AR, there are 26 A-Z Alphabet Flashcards. In AR Mag, there are 5 AR Magazine Pages, and in Eng. Poem, there are 4 English Poems.

The system only works with flashcards that are already trained and recognized. It cannot identify or work with unknown images that are not part of the pre-trained set. Also, the image target data is stored directly on the device, not in the cloud, which might limit how the app can be updated or accessed from different devices. There is no

user account registration in the app, meaning that all users have the same access to the content without any restrictions or personalized settings.

4.4 Further Extension

This “AR-KID” system has great potential for future improvements and extensions to enhance its educational value. One possible extension is to expand the content library by adding more flashcards, magazine pages, and English poems. This would provide a broader range of topics and learning materials for children.

Another important extension could be the integration of a feature that allows the app to recognize and respond to new, untrained images. By incorporating machine learning algorithms, the system could become more adaptable and capable of identifying a wider variety of images, enhancing its functionality and making it more versatile.

Moving the image target data to a cloud-based system would also be a valuable upgrade. This would allow for easier updates, synchronization across multiple devices, and potentially faster image recognition, making the app more efficient and user-friendly.

Adding user account registration and content access controls could also be considered. This would enable personalized learning experiences, where users could save their progress, access specific content based on their learning level, and have their data protected with secure login credentials.

Finally, the system could be extended to include more interactive features, such as quizzes, games, and progress tracking. These additions would make the learning experience even more engaging and help reinforce the educational content in a fun and interactive way.

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